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(54) **SELF-GRIPPING HERNIA PROSTHESIS**

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CPC **A61F 2/0063** (2013.01); **A61F 2210/0004**
(2013.01); **A61F 2220/0016** (2013.01); **A61F**
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(2013.01)

(58) **Field of Classification Search**

CPC **A61F 2/0063**; **D10B 2509/08**; **A61B**
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(56) **References Cited**

U.S. PATENT DOCUMENTS

3,054,406 A 9/1962 Usher
3,124,136 A 3/1964 Usher

(Continued)

FOREIGN PATENT DOCUMENTS

CN 103705276 A 4/2014
CN 103874465 A 6/2014

(Continued)

OTHER PUBLICATIONS

International Preliminary Report on Patentability mailed Apr. 30,
2020, in connection with International Application No. PCT/US2018/
056696.

(Continued)

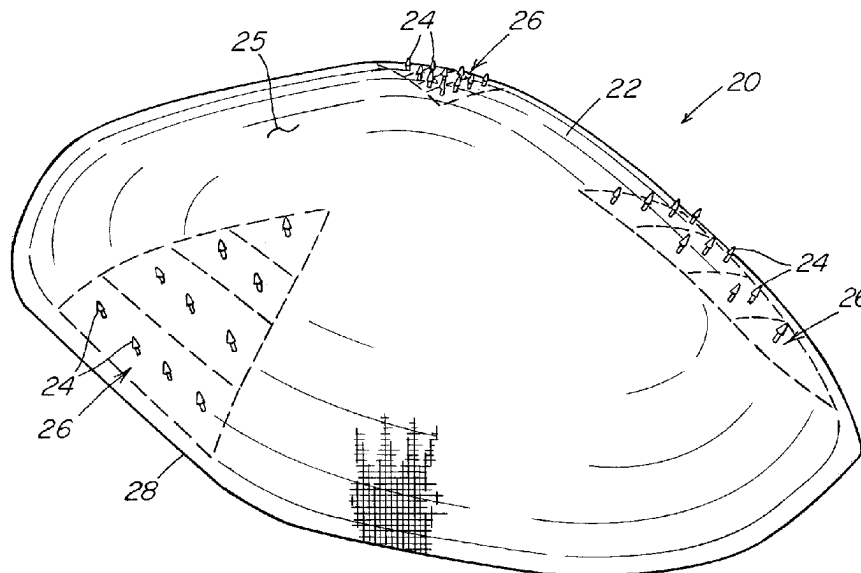
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(57) **ABSTRACT**

A self-gripping hernia prosthesis including a layer of repair
fabric and a plurality of tissue grips protruding from a
surface of the repair fabric. The grip may be fabricated
independent of the repair fabric and subsequently attached to
the layer of fabric. A backing layer may be employed to
secure each grip to the repair fabric. Each base may include
a base that is located between the repair fabric and the
backing layer. Alternatively, the base may be attached
directly to the backing layer. The tissue grips may be
configured to minimize entanglement with the repair fabric.

10 Claims, 15 Drawing Sheets



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References Cited

U.S. PATENT DOCUMENTS

3,320,649	A	5/1967	Naimer	6,991,643	B2	1/2006	Saadat
3,494,006	A	2/1970	Burmlik	7,060,103	B2	6/2006	Carr, Jr. et al.
3,629,930	A	12/1971	Brumlik	7,083,648	B2	8/2006	Yu et al.
RE27,725	E	8/1973	Brumlik	7,094,261	B2	8/2006	Zotti et al.
3,981,051	A	9/1976	Brumlik	7,112,209	B2	9/2006	Ramshaw et al.
4,169,303	A	10/1979	Lemelson	7,156,862	B2	1/2007	Jacobs et al.
4,338,800	A	7/1982	Matsuda	7,172,615	B2	2/2007	Morriss et al.
4,347,847	A	9/1982	Usher	7,214,236	B2	5/2007	Kieturakis et al.
4,452,245	A	6/1984	Usher	7,255,675	B2	8/2007	Gertner et al.
4,463,486	A	8/1984	Matsuda	7,273,497	B2	9/2007	Ferree
4,520,821	A	6/1985	Schmidt et al.	7,331,199	B2	2/2008	Ory et al.
4,653,486	A	3/1987	Coker	7,351,250	B2	4/2008	Zamierowski
4,854,136	A	8/1989	Coslovi et al.	7,381,211	B2	6/2008	Zamierowski
4,976,728	A	12/1990	Willert et al.	7,404,819	B1	7/2008	Darios et al.
5,019,096	A	5/1991	Fox, Jr. et al.	7,410,495	B2	8/2008	Zamierowski
5,032,122	A	7/1991	Noel et al.	7,413,569	B2	8/2008	Sogaard-Andersen
5,116,357	A	5/1992	Eberbach	7,413,570	B2	8/2008	Zamierowski
5,147,374	A	9/1992	Fernandez	7,413,571	B2	8/2008	Zamierowski
5,176,692	A	1/1993	Wilk et al.	7,497,864	B2	3/2009	Bonutti
5,234,488	A	8/1993	Ichikawa et al.	7,510,566	B2	3/2009	Jacobs et al.
5,254,127	A	10/1993	Wholey et al.	7,517,313	B2	4/2009	Thierfelder et al.
5,254,133	A	10/1993	Seid	7,544,213	B2	6/2009	Adams
5,258,000	A	11/1993	Gianturco	7,566,337	B2	7/2009	Sogaard-Andersen et al.
5,330,445	A	7/1994	Haaga	7,658,749	B2	2/2010	Wittmann
5,356,432	A	10/1994	Rutkow et al.	7,662,169	B2	2/2010	Wittmann
5,378,522	A	1/1995	Lagomarsino	7,722,528	B2	5/2010	Arnal et al.
5,383,897	A	1/1995	Wholey	7,905,825	B2	3/2011	Arnal et al.
5,569,273	A	10/1996	Titone et al.	8,047,981	B2	11/2011	Rao et al.
5,634,931	A	6/1997	Kugel	8,062,331	B2	11/2011	Zamierowski
5,643,295	A	7/1997	Yoon	8,070,773	B2	12/2011	Zamierowski
5,716,408	A	2/1998	Eldridge et al.	8,123,781	B2	2/2012	Zamierowski
5,716,409	A	2/1998	Debbas	8,128,655	B2	3/2012	Wittmann
5,761,775	A	6/1998	Legome et al.	8,157,722	B2	4/2012	Arnal et al.
5,769,864	A	6/1998	Kugel	8,188,834	B2	5/2012	Berres
5,836,961	A	11/1998	Kieturakis et al.	8,216,273	B1	7/2012	Goralchouk et al.
5,906,617	A	5/1999	Meislin	8,273,105	B2	9/2012	Cohen et al.
5,916,225	A	6/1999	Kugel	8,435,307	B2	5/2013	Paul
6,042,534	A	3/2000	Gellman et al.	8,454,653	B2	6/2013	Hadba et al.
6,090,116	A	7/2000	D'Aversa et al.	8,460,338	B2	6/2013	Goralchouk et al.
6,176,863	B1	1/2001	Kugel et al.	8,469,996	B2	6/2013	Wittmann
6,224,616	B1	5/2001	Kugel	8,500,759	B2	8/2013	Koyfman et al.
6,241,747	B1	6/2001	Ruff	8,562,644	B2	10/2013	Yuan et al.
6,280,453	B1	8/2001	Kugel et al.	8,579,924	B2	11/2013	Stopek et al.
6,287,316	B1	9/2001	Agarwal et al.	8,585,576	B2	11/2013	Arnal et al.
6,290,708	B1	9/2001	Kugel et al.	8,585,721	B2	11/2013	Kirsch
6,312,442	B1	11/2001	Kieturakis et al.	8,615,856	B1	12/2013	Gelbart
6,319,264	B1	11/2001	Tormala et al.	8,618,096	B2	12/2013	Renslo et al.
6,383,201	B1	5/2002	Dong	8,632,567	B2	1/2014	Cohen et al.
6,425,924	B1	7/2002	Rousseau	8,632,839	B2	1/2014	Stopek et al.
6,468,314	B2 *	10/2002	Schwartz A61F 2/30749 623/23.72	8,690,960	B2	4/2014	Hotter et al.
6,485,503	B2	11/2002	Jacobs et al.	8,739,389	B2	6/2014	Cohen et al.
6,530,933	B1	3/2003	Yeung et al.	8,747,436	B2	6/2014	Nawrocki et al.
6,565,590	B2	5/2003	Kieturakis et al.	8,793,863	B2	8/2014	Hunter et al.
6,592,515	B2	7/2003	Thierfelder et al.	8,821,539	B2	9/2014	Rousseau
6,596,002	B2	7/2003	Therin et al.	8,888,810	B2	11/2014	Hadba et al.
6,596,304	B1	7/2003	Bayon et al.	8,906,046	B2	12/2014	Anderson
6,599,318	B1	7/2003	Gabbay	8,932,327	B2	1/2015	Kosa et al.
6,645,226	B1	11/2003	Jacobs et al.	8,932,329	B2	1/2015	Hadba et al.
6,679,900	B2	1/2004	Kieturakis et al.	8,961,850	B2	2/2015	Wood et al.
6,695,856	B2	2/2004	Kieturakis et al.	8,968,418	B2	3/2015	Paul
6,740,122	B1 *	5/2004	Pajotin A61F 2/0063 606/151	9,034,011	B2	5/2015	Kirsch et al.
6,746,458	B1	6/2004	Cloud	9,050,082	B2	6/2015	Cohen et al.
6,758,853	B2	7/2004	Kieturakis et al.	9,060,767	B2	6/2015	Bonutti
6,790,213	B2	9/2004	Cherok et al.	9,186,235	B2	11/2015	Ory et al.
6,837,078	B1	1/2005	Rock et al.	9,186,236	B2	11/2015	Paul
6,884,212	B2	4/2005	Thierfelder et al.	9,295,538	B2	3/2016	Koyfman et al.
6,893,452	B2	5/2005	Jacobs	9,308,068	B2	4/2016	Spinnler et al.
				9,358,002	B2	6/2016	Kirsch et al.
				9,398,943	B2	7/2016	Criscuolo et al.
				9,439,749	B2	9/2016	Lecuire et al.
				9,445,883	B2	9/2016	Lecuire et al.
				9,452,038	B2	9/2016	Amal et al.
				9,480,547	B2	11/2016	Paul
				9,510,927	B2	12/2016	Simons
				9,510,931	B2	12/2016	Hotter et al.
				9,598,801	B2	3/2017	Lecuire
				9,629,703	B2	4/2017	Paul
				9,677,208	B2	6/2017	Meneghin
				9,713,467	B2	7/2017	Cohen et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,744,019 B2	8/2017	Montanari et al.	2009/0216075 A1	8/2009	Bell et al.
9,744,919 B2	8/2017	Schutz	2009/0216338 A1	8/2009	Gingras et al.
9,750,595 B2	9/2017	Thomas et al.	2009/0228021 A1	9/2009	Leung
9,788,832 B2	10/2017	Hadba et al.	2009/0240267 A1	9/2009	Crawley et al.
9,801,703 B2	10/2017	Rao et al.	2009/0240343 A1	9/2009	Adams
9,949,815 B2	4/2018	Priewe	2009/0248067 A1	10/2009	Maiorino
9,962,162 B2	5/2018	Bonutti	2009/0248070 A1	10/2009	Kosa et al.
9,962,250 B2	5/2018	Priewe et al.	2009/0299407 A1	12/2009	Yuan et al.
10,052,184 B2	8/2018	Deichmann et al.	2009/0306681 A1	12/2009	Del Nido et al.
10,058,326 B2	8/2018	Kirsch et al.	2010/0191044 A1	7/2010	Gobron et al.
10,136,984 B2 *	11/2018	Fischer	2010/0211097 A1	8/2010	Hadba et al.
2001/0010023 A1 *	7/2001	Schwartz	2010/0255447 A1	10/2010	Biris et al.
			2010/0268272 A1	10/2010	Kirsch et al.
			2010/0274283 A1	10/2010	Kirsch et al.
			2010/0292719 A1	11/2010	Ducharme
			2011/0028995 A1	2/2011	Miraki et al.
			2011/0077457 A1 *	3/2011	Deitch
					A61F 2/0045 600/37
2002/0022861 A1	2/2002	Jacobs et al.	2011/0082478 A1	4/2011	Glick et al.
2002/0058966 A1	5/2002	Tormala et al.	2011/0130774 A1	6/2011	Criscuolo et al.
2002/0077661 A1 *	6/2002	Saadat	2011/0160527 A1	6/2011	Khamis et al.
			2011/0172760 A1	7/2011	Anderson
			2011/0238094 A1	9/2011	Thomas et al.
2002/0103494 A1	8/2002	Pacey	2011/0276090 A1	11/2011	Berndt et al.
2002/0169452 A1	11/2002	Tormala et al.	2011/0282365 A1	11/2011	Hadba et al.
2002/0188170 A1	12/2002	Santamore et al.	2011/0282386 A1	11/2011	Friedrich et al.
2003/0036801 A1	2/2003	Schwartz et al.	2011/0288565 A1	11/2011	Kubiak et al.
2003/0069602 A1	4/2003	Jacobs et al.	2011/0288566 A1	11/2011	Kubiak
2003/0074021 A1 *	4/2003	Morriss	2012/0010637 A1	1/2012	Stopek et al.
			2012/0027804 A1	2/2012	Odermatt et al.
			2012/0029538 A1 *	2/2012	Reeser
					A61B 17/068 606/151
2003/0105477 A1 *	6/2003	Schwartz	2012/0053399 A1	3/2012	Rao et al.
			2012/0082712 A1	4/2012	Stopek et al.
			2012/0094004 A1	4/2012	Stopek et al.
			2012/0094005 A1	4/2012	Stopek et al.
			2012/0165937 A1	6/2012	Montanari et al.
			2012/0172926 A1	7/2012	Hotter
			2012/0179175 A1	7/2012	Hammell
			2012/0197415 A1 *	8/2012	Montanari
					A61L 27/56 623/23.74
2003/0187516 A1 *	10/2003	Amid	2012/0310278 A1	12/2012	Marczyk et al.
			2012/0330093 A1	12/2012	Odermatt et al.
			2013/0018393 A1	1/2013	Bengston
			2013/0102959 A1	4/2013	Stopek et al.
			2013/0103060 A1	4/2013	Stopek et al.
			2013/0138124 A1	5/2013	Criscuolo et al.
			2013/0158571 A1 *	6/2013	Meneghin
					D04B 21/12 606/151
2004/0059356 A1	3/2004	Gingras	2013/0158572 A1	6/2013	Meneghin et al.
2004/0138683 A1 *	7/2004	Shelton	2013/0172915 A1	7/2013	Thomas et al.
			2013/0172927 A1 *	7/2013	Natarajan
					A61F 2/0077 264/319
2004/0138705 A1	7/2004	Heino et al.	2013/0184722 A1	7/2013	Stopek et al.
2004/0204723 A1	10/2004	Kayan	2013/0190785 A1	7/2013	Koyfman et al.
2004/0215219 A1	10/2004	Eldridge et al.	2013/0204277 A1	8/2013	Fabry et al.
2004/0260340 A1	12/2004	Jacobs et al.	2013/0218125 A1	8/2013	Stopek et al.
2005/0021058 A1	1/2005	Negro	2013/0231688 A1	9/2013	Paul
2005/0085924 A1	4/2005	Darois et al.	2013/0231702 A1	9/2013	Hadba et al.
2005/0113858 A1	5/2005	Deutsch	2013/0261542 A1	10/2013	Elachchabi et al.
2005/0119694 A1	6/2005	Jacobs et al.	2013/0261594 A1	10/2013	Stopek et al.
2005/0216042 A1	9/2005	Gertner	2013/0331792 A1	12/2013	Karp et al.
2005/0222591 A1	10/2005	Gingras et al.	2014/0046348 A1 *	2/2014	Soltanian
2005/0228408 A1	10/2005	Fricke et al.			A61B 17/064 606/151
2006/0089672 A1	4/2006	Martinek	2014/0066958 A1	3/2014	Priewe
2006/0141012 A1	6/2006	Gingras	2014/0081295 A1	3/2014	Lau et al.
2006/0173471 A1	8/2006	Carr et al.	2014/0094829 A1	4/2014	Kostrzewski
2006/0200062 A1	9/2006	Saadat	2014/0094830 A1 *	4/2014	Sargeant
2006/0205995 A1	9/2006	Browning			A61F 2/0063 606/151
2006/0253132 A1	11/2006	Evans et al.	2014/0094831 A1	4/2014	Thomas et al.
2006/0253203 A1	11/2006	Alvarado	2014/0142368 A1	5/2014	Arnal et al.
2006/0282103 A1	12/2006	Fricke et al.	2014/0142520 A1	5/2014	Stopek et al.
2007/0027358 A1	2/2007	Gertner et al.	2014/0148827 A1 *	5/2014	Odermatt
2007/0068538 A1	3/2007	Anderson et al.			A61B 17/0057 606/151
2007/0083229 A1	4/2007	Deutsch	2014/0222164 A1	8/2014	Hotter et al.
2007/0123936 A1	5/2007	Goldin et al.	2014/0224094 A1	8/2014	Cohen et al.
2007/0260268 A1	11/2007	Bartee et al.	2014/0228867 A1 *	8/2014	Thomas
2007/0265704 A1 *	11/2007	Mayer			A61F 2/0063 264/293
2008/0009902 A1	1/2008	Hunter et al.			
2008/0033461 A1	2/2008	Koeckerling et al.			
2008/0113001 A1 *	5/2008	Herweck			
2008/0195231 A1	8/2008	Ory et al.			
2008/0200950 A1	8/2008	Wohlert			
2008/0206305 A1	8/2008	Herweck et al.			
2008/0208079 A1	8/2008	Hein et al.			
2008/0255593 A1	10/2008	St-Germain			
2008/0281357 A1	11/2008	Sung et al.			
2009/0024147 A1	1/2009	Ralph et al.			
2009/0082792 A1	3/2009	Koyfman et al.			
2009/0112236 A1	4/2009	Stopek			
2009/0163938 A1	6/2009	Bonutti			
2009/0187197 A1	7/2009	Roerber et al.			
2009/0192532 A1	7/2009	Spinnler et al.			
2009/0204129 A1	8/2009	Fronio			

FOREIGN PATENT DOCUMENTS

DE 198 32 634 A1 1/2000

2014/0257348	A1	9/2014	Priewe et al.	
2014/0257517	A1	9/2014	Deichmann et al.	
2015/0012089	A1	1/2015	Shetty et al.	
2015/0040534	A1	2/2015	Hadba et al.	
2015/0057762	A1 *	2/2015	Harms	A61F 2/02 623/23.74
2015/0080918	A1 *	3/2015	Lecuire	D04B 21/12 606/151
2015/0119933	A1	4/2015	Kosa et al.	
2015/0133998	A1	5/2015	Cohen et al.	
2015/0196378	A1 *	7/2015	Mayer	A61B 17/8605 606/151
2015/0245837	A1	9/2015	Bonutti	
2015/0313700	A1 *	11/2015	Rizk	A61F 2/0063 606/151
2015/0315729	A1 *	11/2015	Simons	D04B 21/12 66/170
2016/0022405	A1	1/2016	Spinnler et al.	
2016/0135969	A1 *	5/2016	Anderson	A61B 17/0644 623/1.18
2016/0262749	A1	9/2016	Kirsch et al.	
2016/0374791	A1	12/2016	Lecuire et al.	
2016/0374792	A1	12/2016	Lecuire et al.	
2017/0105724	A1 *	4/2017	Limem	A61B 17/064
2017/0189158	A1	7/2017	Lecuire	
2017/0202654	A1	7/2017	Paul	
2017/0209251	A1 *	7/2017	Francois	A61F 2/0063
2017/0231740	A1 *	8/2017	Peery	A61B 17/064 606/151
2017/0296319	A1 *	10/2017	Peery	A61B 17/068
2017/0319203	A1	11/2017	Cohen et al.	
2018/0008390	A1	1/2018	Montanari et al.	
2018/0161141	A1	6/2018	Priewe et al.	
2018/0235740	A1	8/2018	Deichmann et al.	
2018/0250007	A1	9/2018	Bonutti	
2018/0318057	A1 *	11/2018	Bailly	A61F 2/0063
2018/0325518	A1 *	11/2018	Tannhauser	A61B 17/07292
2019/0117363	A1	4/2019	Felix et al.	

DE	198 32 634	A1	1/2000
EP	0 276 890	A2	8/1988
EP	1 274 892	B1	11/2009
EP	2 143 836	A1	1/2010
EP	2 229 918	A1	9/2010
EP	2 514 862	A2	10/2012
EP	2 567 715	A1	3/2013
EP	2 712 577	A1	4/2014
EP	2 753 370	B1	1/2016
EP	2 744 934	B1	10/2018
EP	2 798 105	B1	10/2018
EP	2 798 106	B1	10/2018
FR	2 712 177	A1	5/1995
WO	WO 2006/116000	A2	11/2006
WO	WO 2007/120138	A2	10/2007
WO	WO 2009/132284	A2	10/2009
WO	WO 2012/007579	A1	1/2012
WO	WO 2013/026682	A1	2/2013
WO	WO 2013/034641	A1	3/2013
WO	WO 2013/049799	A1	4/2013
WO	WO 2013/098343	A1	7/2013
WO	WO 2013/098345	A1	7/2013
WO	WO 2013/098347	A1	7/2013
WO	WO 2013/139482	A1	9/2013
WO	WO 2014/001432	A1	1/2014
WO	WO 2014/048922	A1	4/2014
WO	WO 2014/139633	A1	9/2014
WO	WO 2014/139635	A1	9/2014
WO	WO 2016/141183	A1	9/2016

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed Feb. 11, 2019, in connection with International Application No. PCT/US2018/056696.

Chinese Office Action mailed Feb. 16, 2022, in connection with Chinese Application No. 201880074867.8.

Chinese Office Action mailed Sep. 9, 2022, in connection with Chinese Application No. 201880074867.8.

* cited by examiner

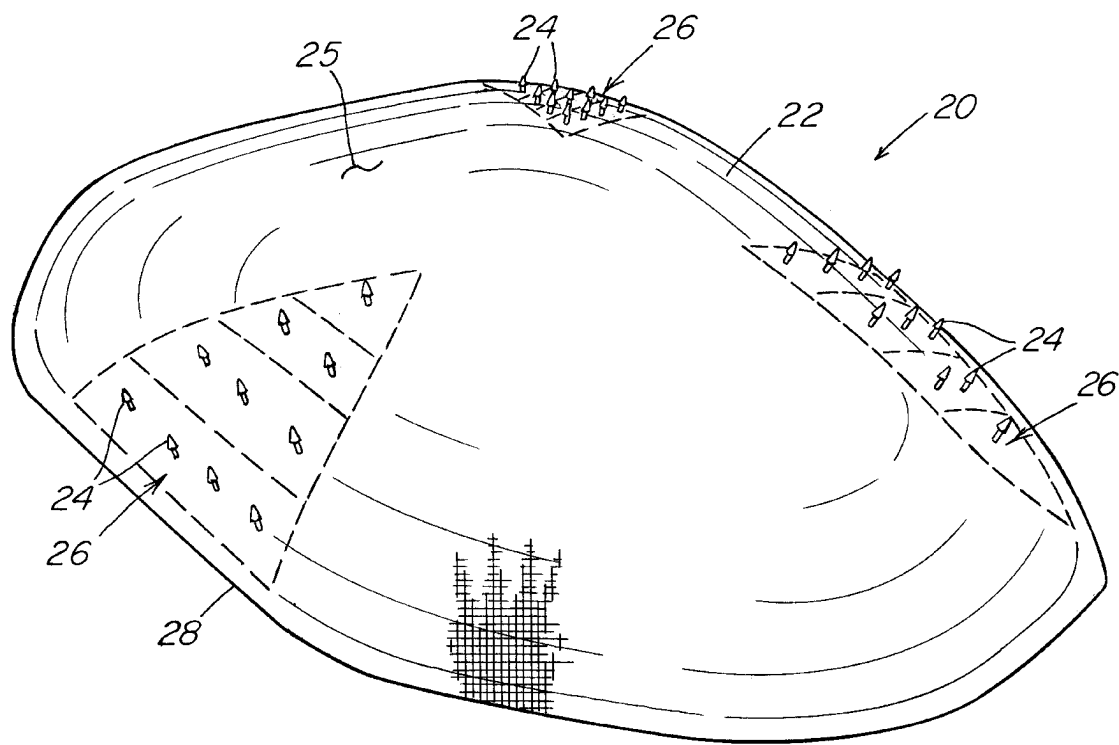


FIG. 1

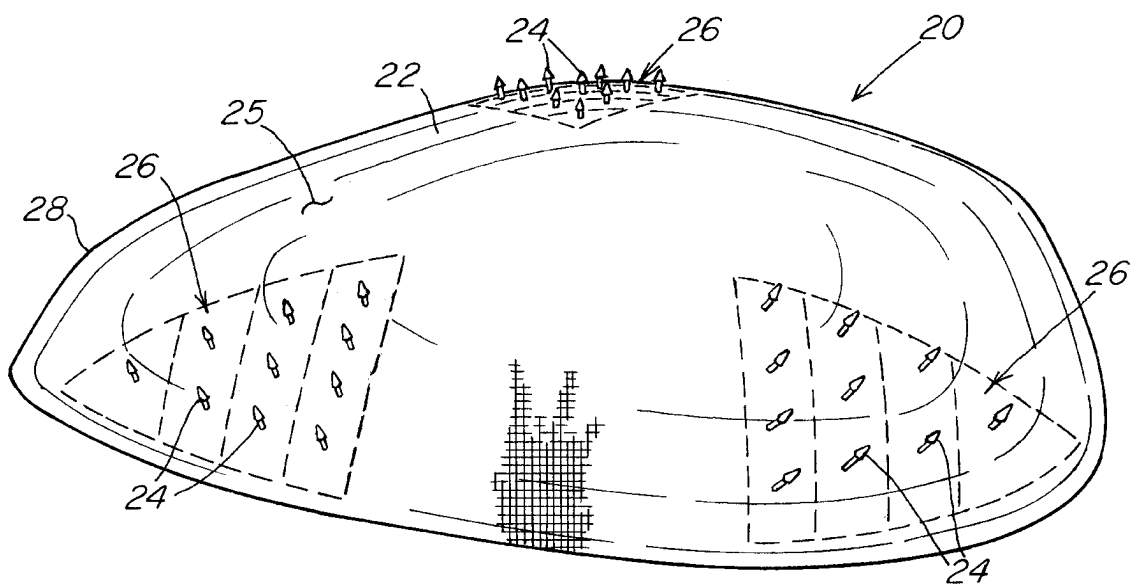


FIG. 2

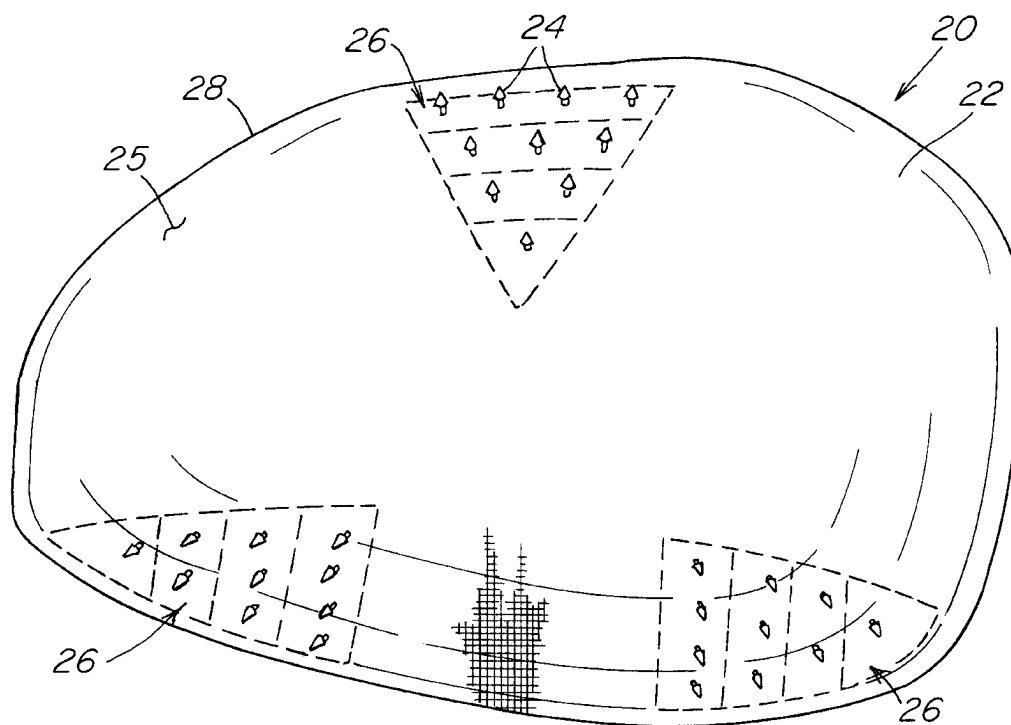


FIG. 3

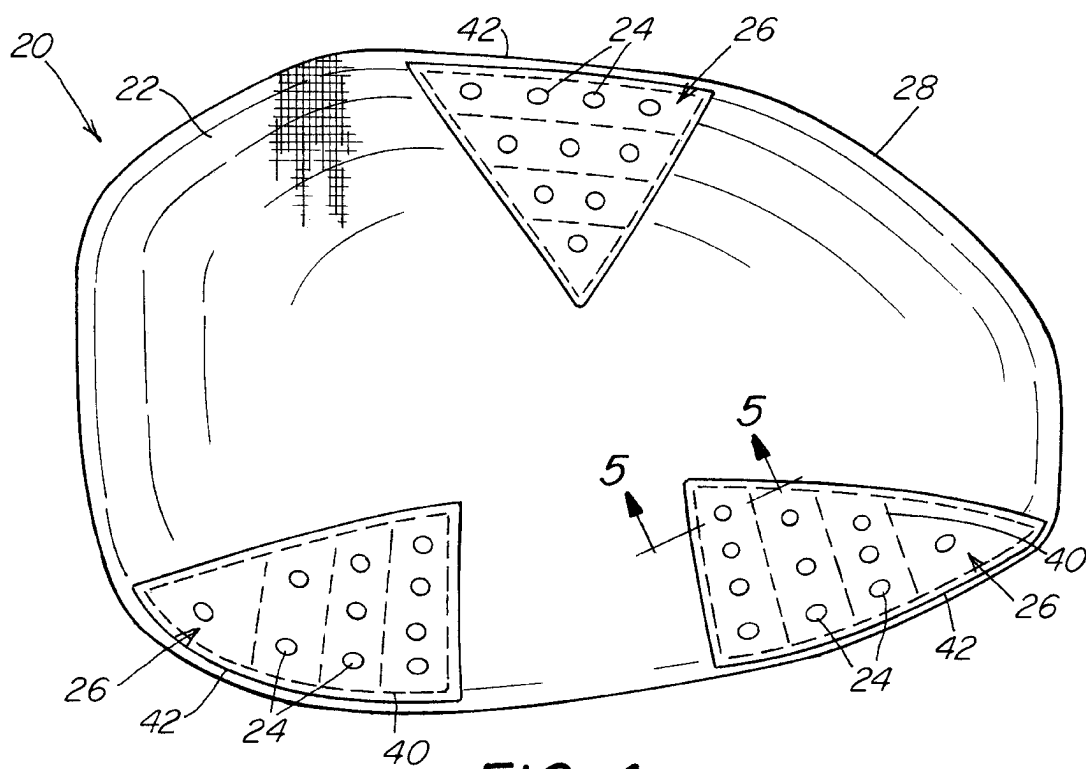


FIG. 4

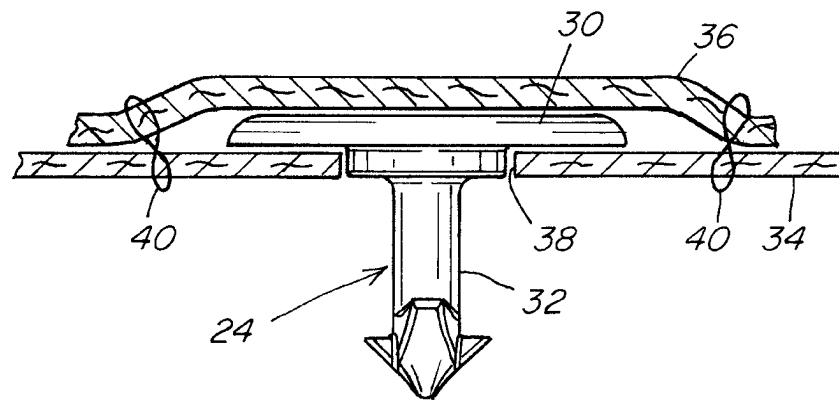


FIG. 5

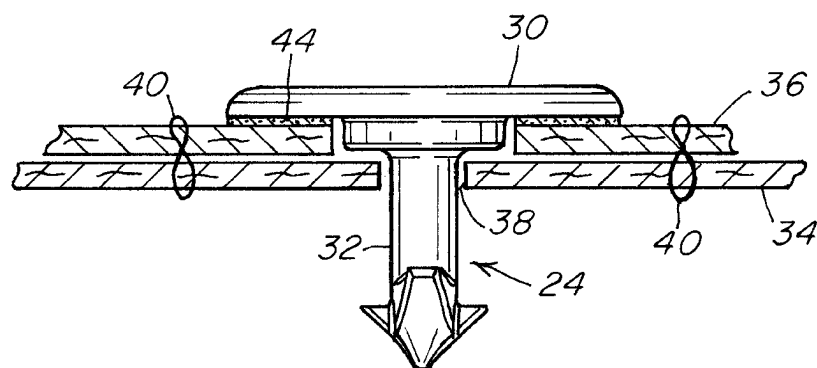


FIG. 6

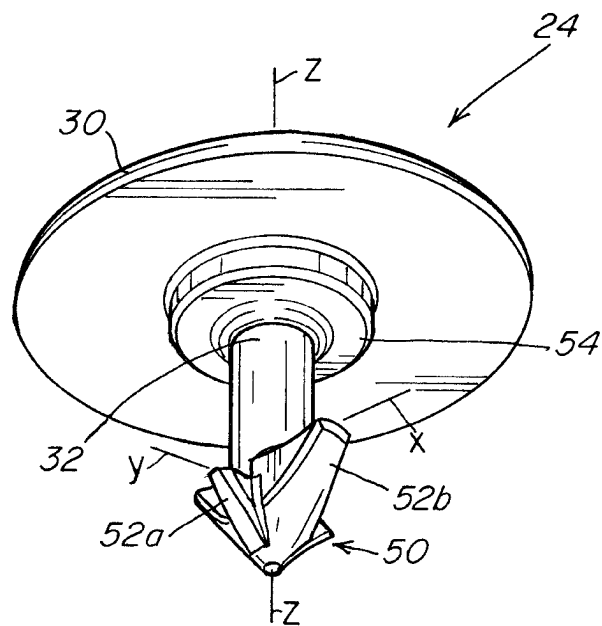


FIG. 7

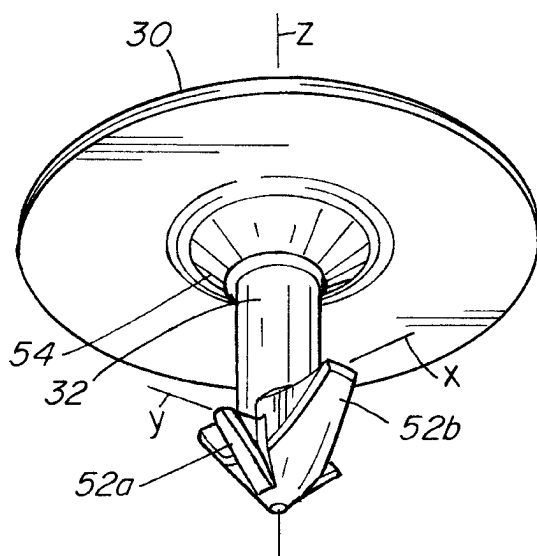


FIG. 8

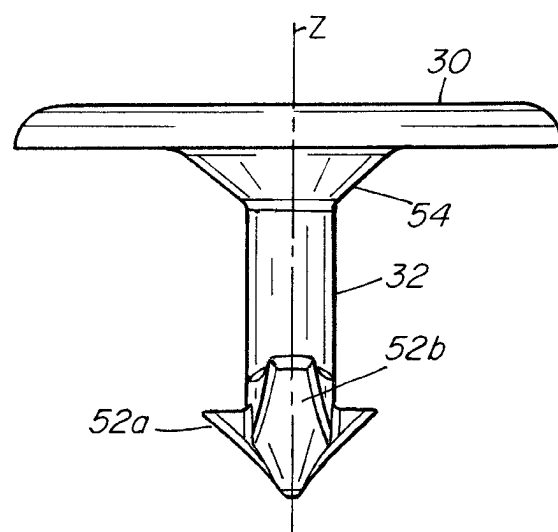


FIG. 9

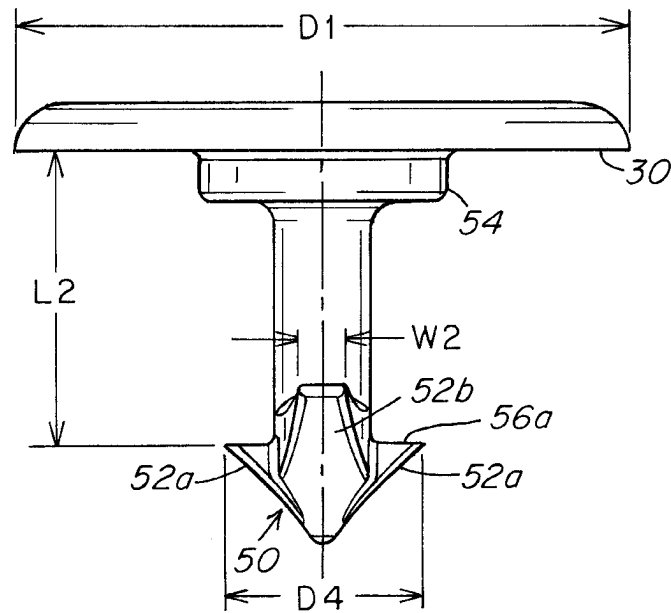


FIG. 10

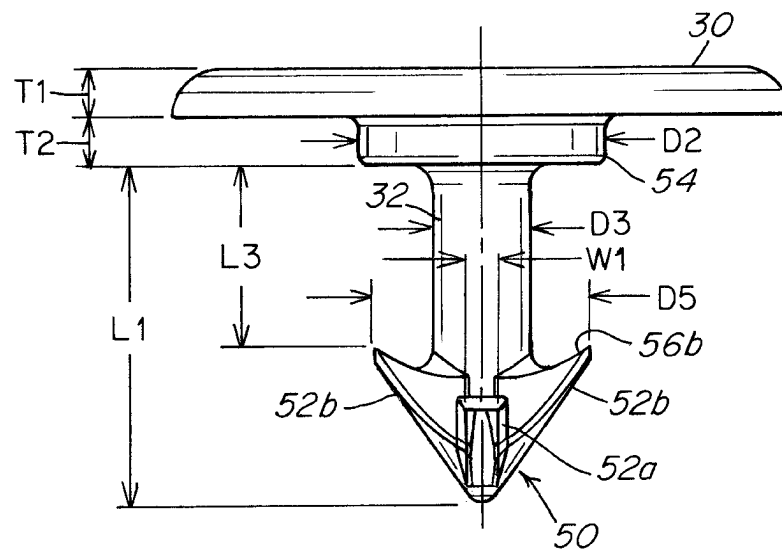


FIG. 11

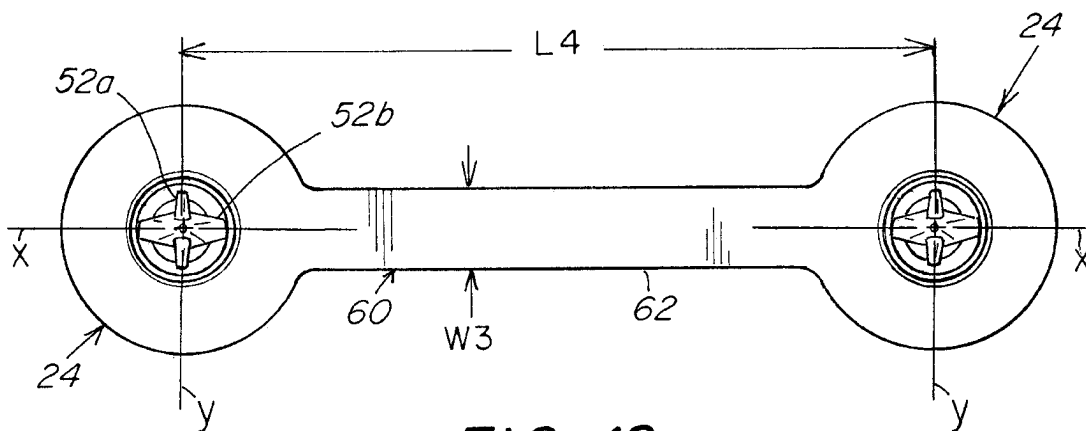


FIG. 12

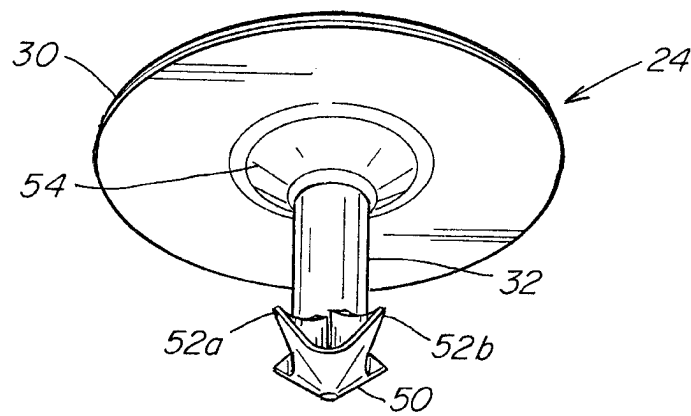


FIG. 13

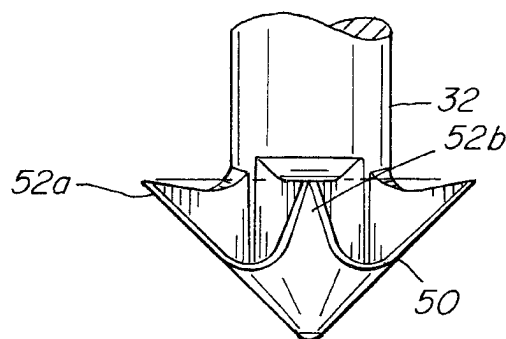


FIG. 14

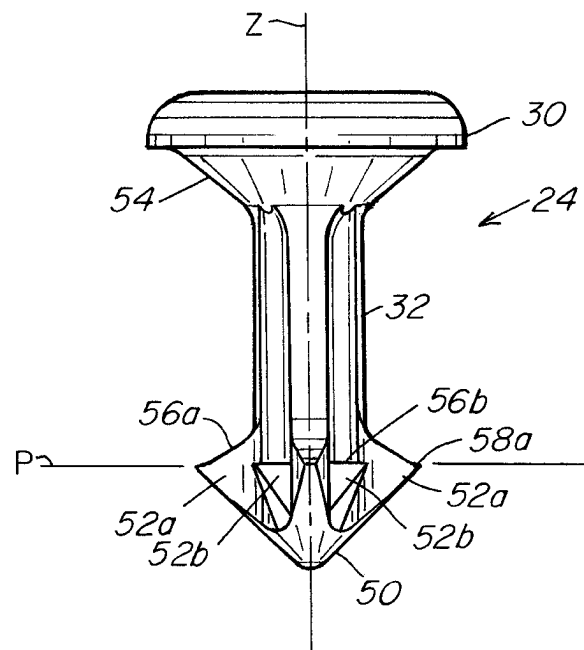


FIG. 15

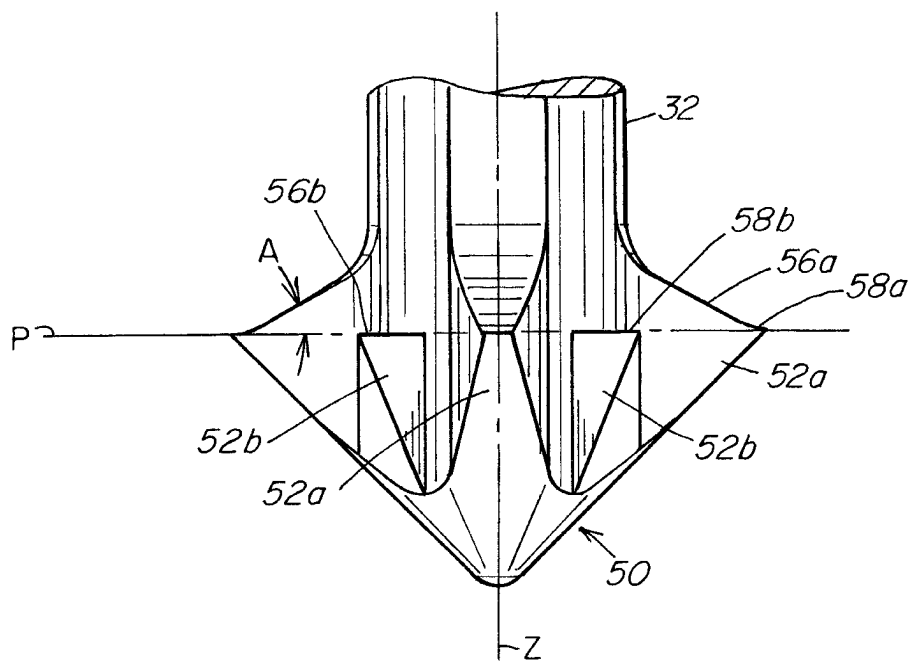


FIG. 16

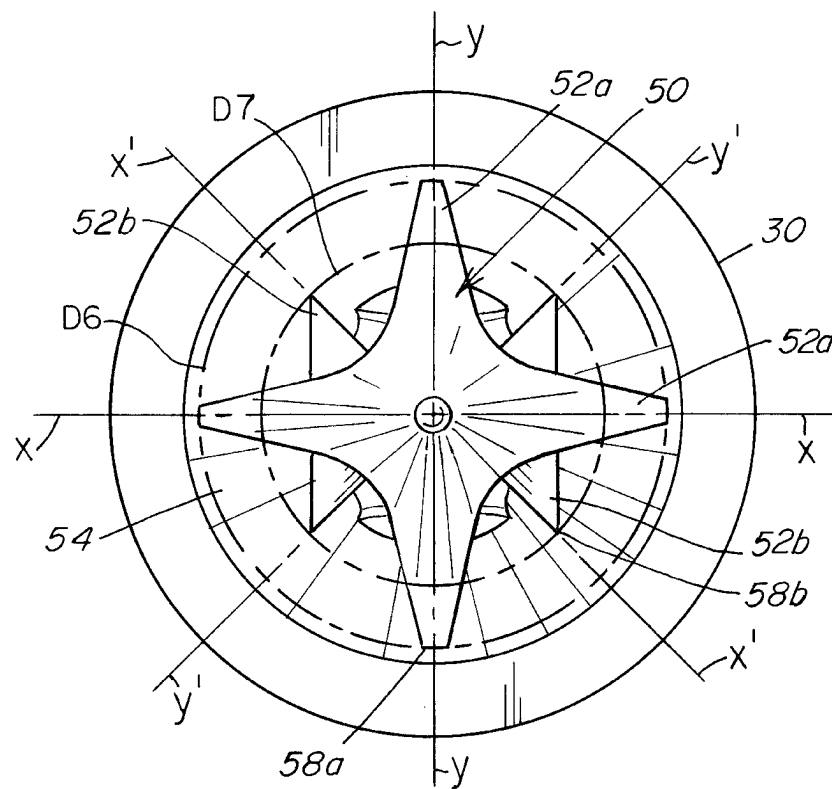


FIG. 17

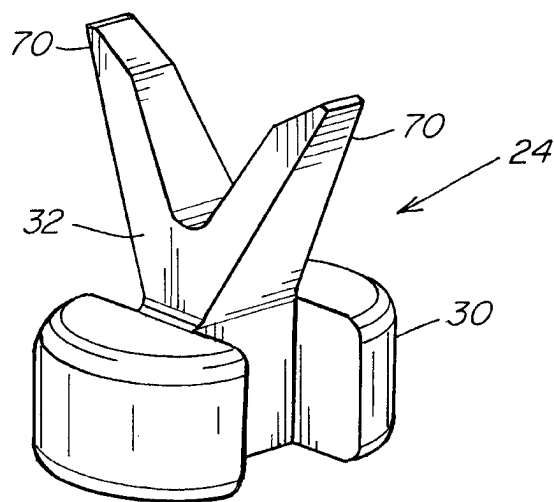


FIG. 18

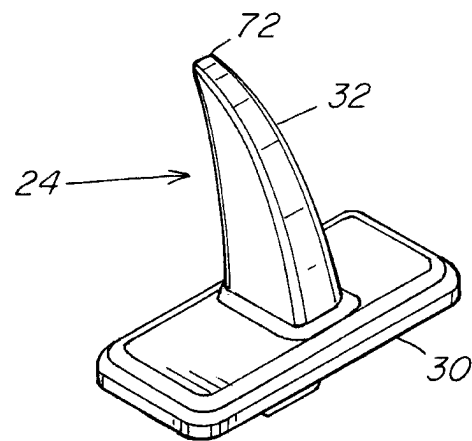


FIG. 19

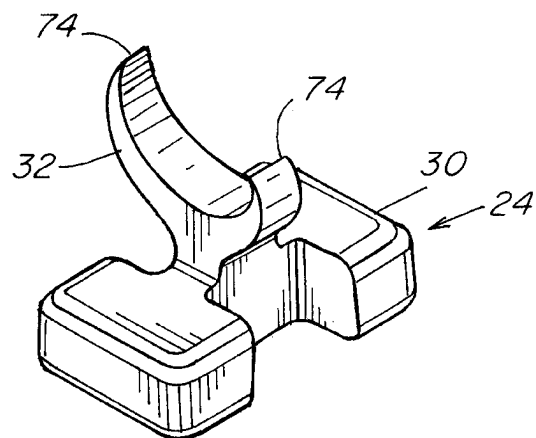


FIG. 20

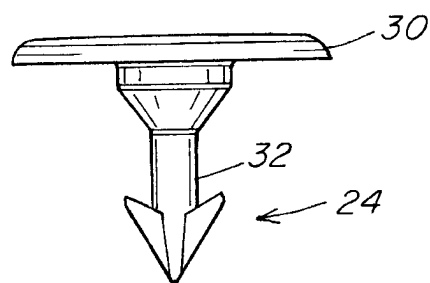


FIG. 21

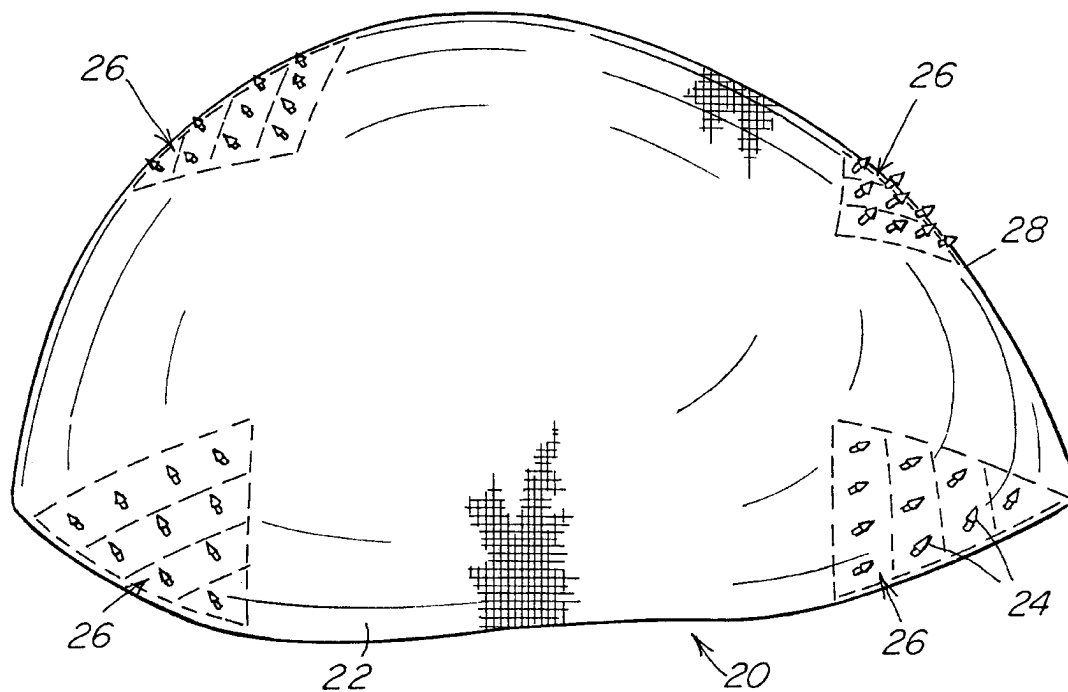


FIG. 22

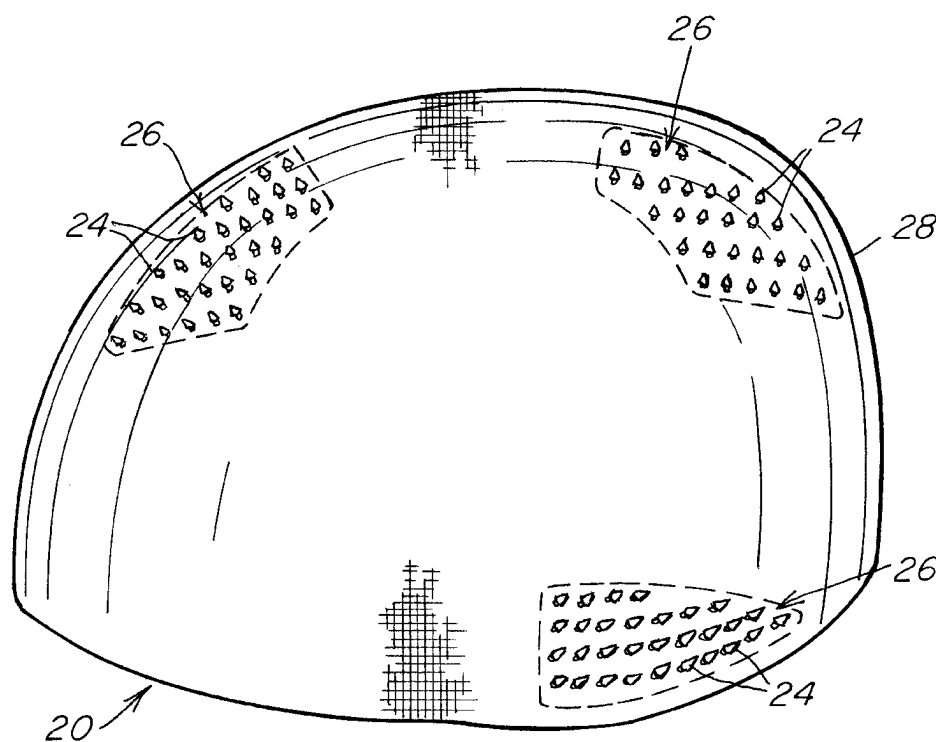


FIG. 23

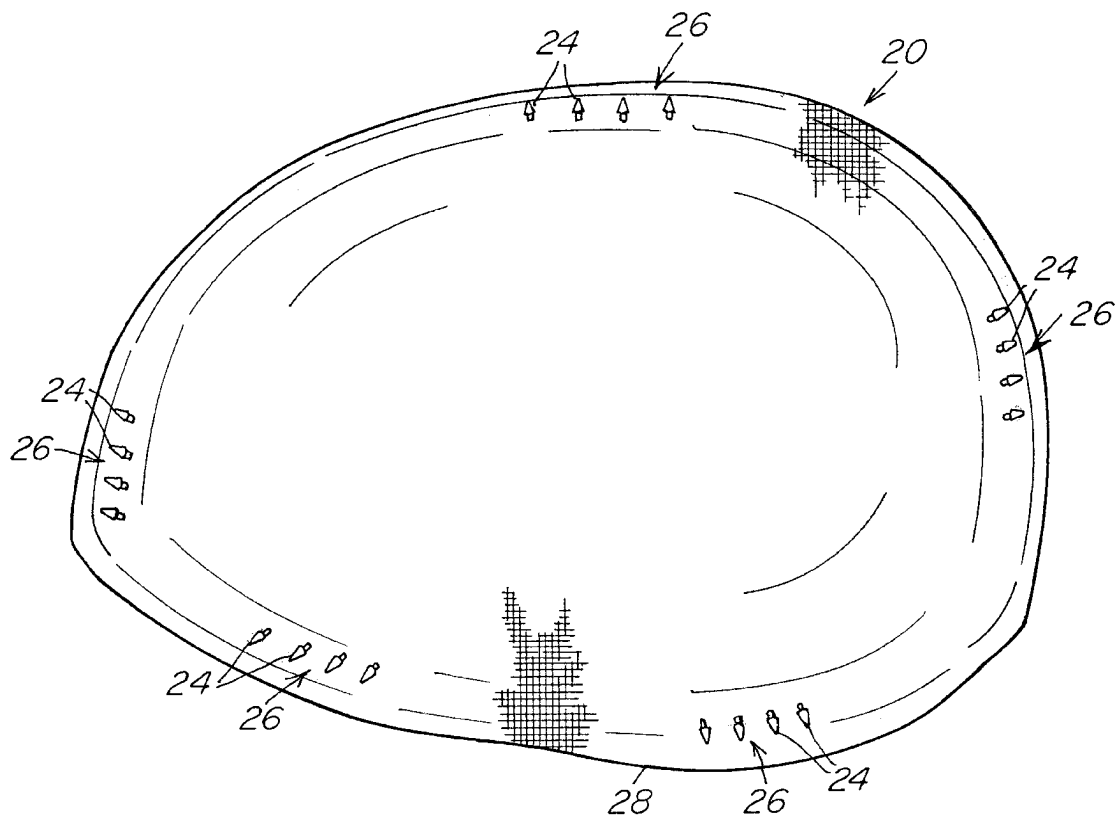


FIG. 24

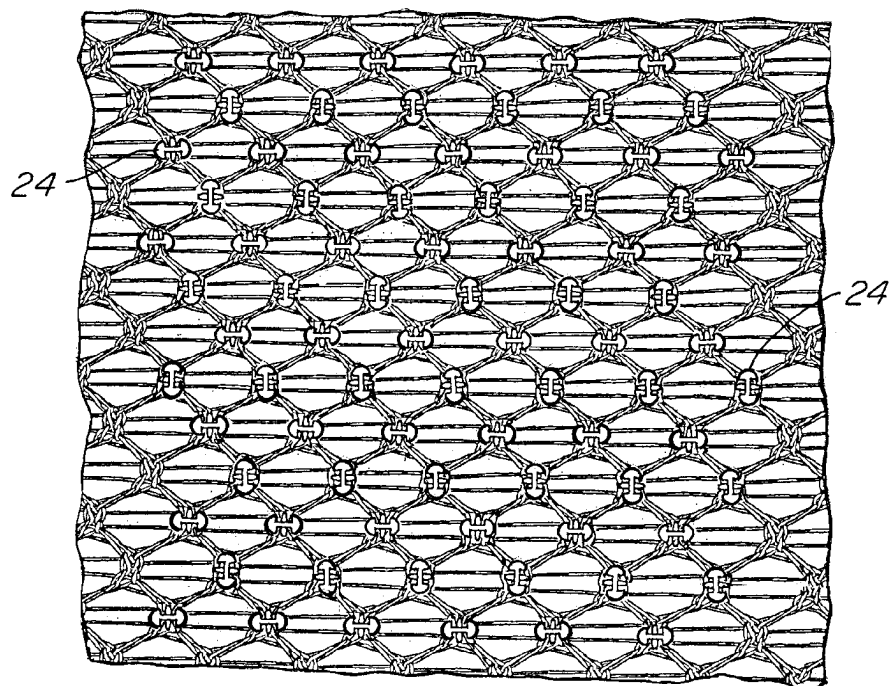


FIG. 25

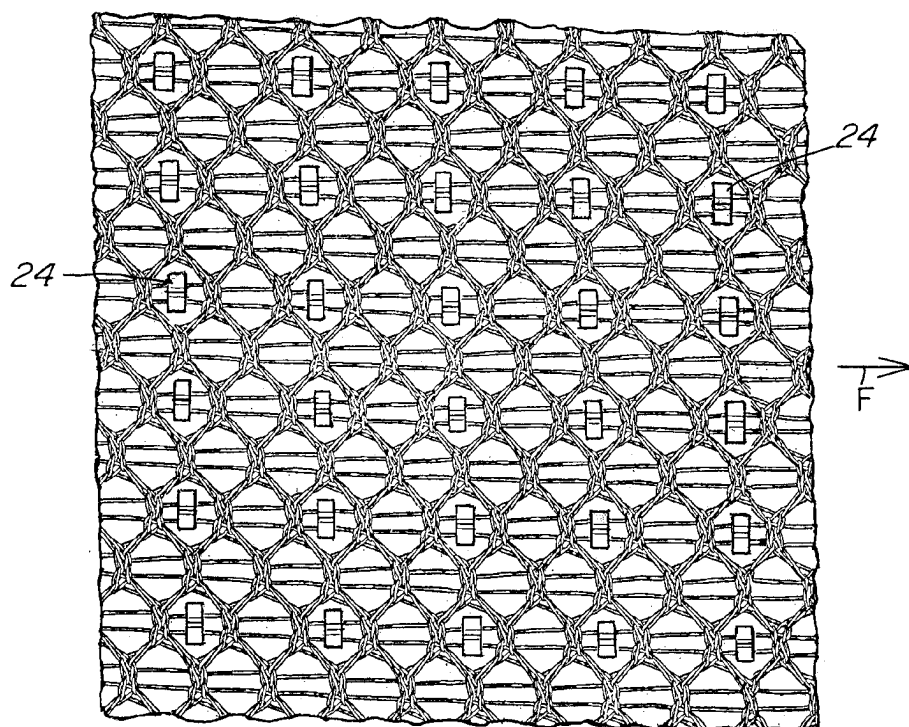


FIG. 26

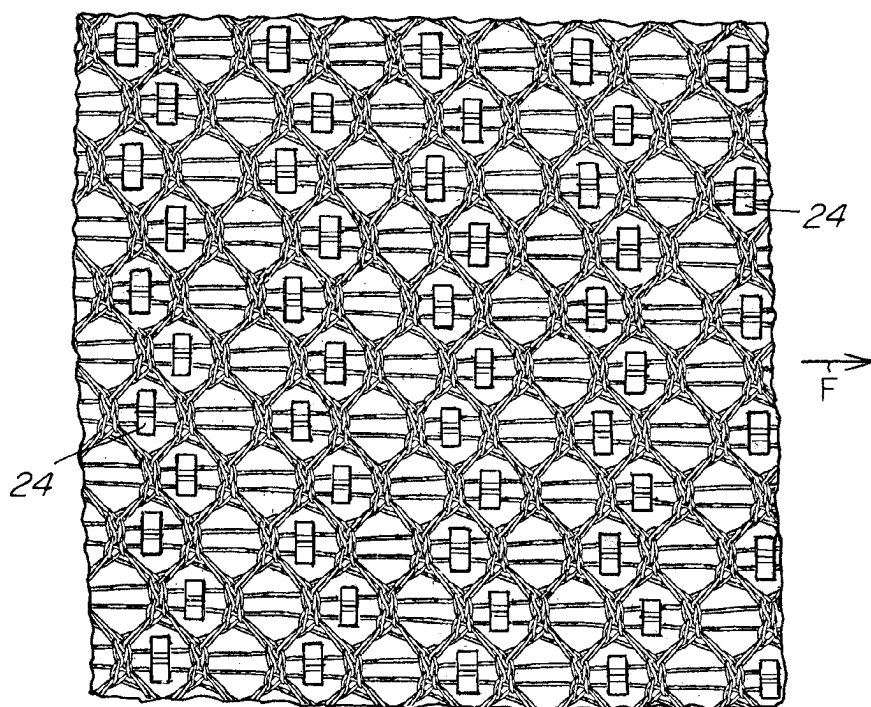


FIG. 27

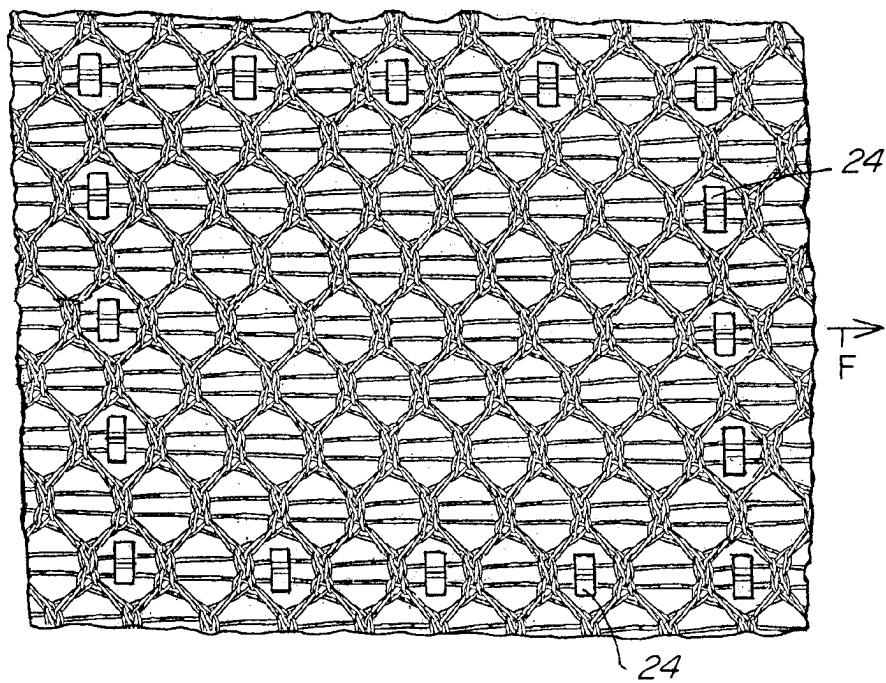


FIG. 28

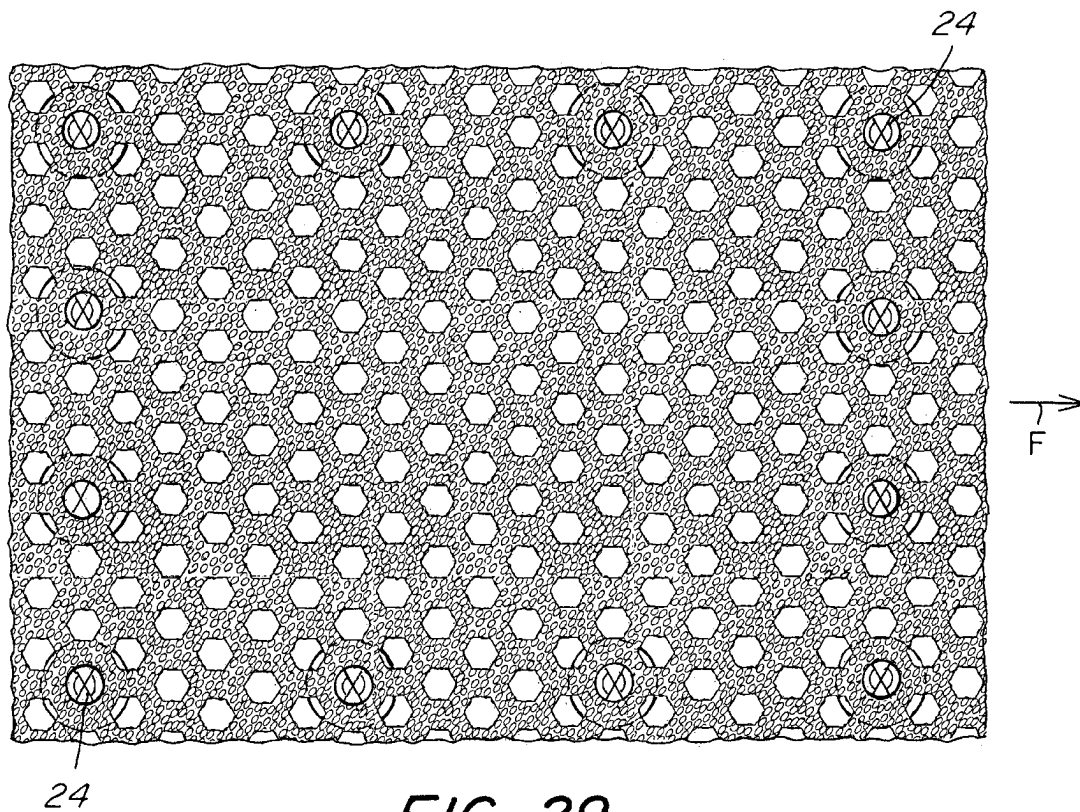


FIG. 29

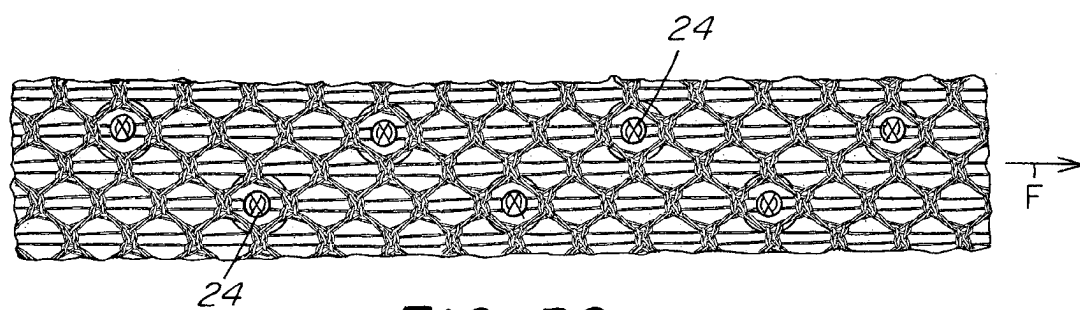


FIG. 30

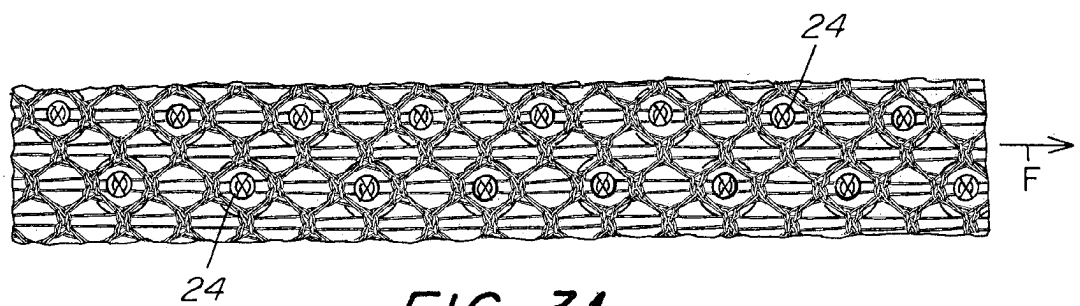


FIG. 31

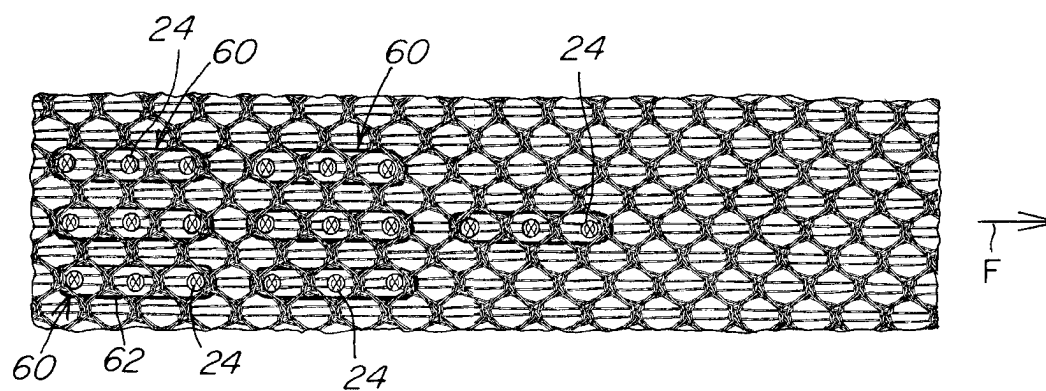


FIG. 32

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SELF-GRIPPING HERNIA PROSTHESIS**RELATED APPLICATIONS**

This application is a divisional of U.S. application Ser. No. 16/165,436, filed Oct. 19, 2018, which claims the benefit of U.S. Provisional Application No. 62/574,525, filed Oct. 19, 2017, each of which is incorporated herein by reference in its entirety.

FIELD

The present invention relates to an implantable prosthesis, and more particularly to a prosthesis for mending defects and weaknesses of soft tissue and muscle walls.

BACKGROUND

A defect in a muscle or tissue wall, such as a hernia, is commonly repaired with an implantable prosthesis that is configured to cover and/or fill the defect.

For some procedures, an implantable repair fabric, such as a mesh fabric, is sutured, stapled, tacked, or otherwise provisionally anchored in place over, under or within the defect. Tissue integration with the fabric, such as tissue ingrowth into and/or along the mesh fabric, eventually completes the repair.

Various repair fabrics are known and used for repairing soft tissue and muscle wall defects. Examples of implantable fabrics that have been successfully used in soft tissue and muscle wall repair include BARD Soft Mesh, BARD Mesh and VISILEX, available from C. R. Bard. Such fabrics are fabricated from polypropylene monofilaments that are knitted into a mesh having pores or interstices that promote tissue ingrowth and integration with the fabric.

For some procedures, it may be desirable to employ an implantable prosthesis configured to fit the shape of the anatomical region of the defect. In some instances, such a prosthesis can be positioned and maintain its position relative to the defect with little or no provisional fixation. Examples of implantable prostheses that have been successfully used in soft tissue and muscle wall repair include 3DMAX Light Mesh and 3DMAX Mesh, available from C. R. Bard. Such prostheses are fabricated from a mesh fabric formed into a curved, 3-dimensional shape that fits the anatomical shape of the defect region, such as the inguinal anatomy. These prostheses have proven useful and have become established in the practice of muscle or tissue wall repair in the inguinothoracic region.

It is an object of the invention to provide a prosthesis for mending defects in soft tissue and muscle walls.

SUMMARY

The present invention relates to an implantable prosthesis for mending an anatomical defect, such as a tissue or muscle wall defect, including a groin defect.

According to one aspect, an implantable prosthesis comprises a first layer of biologically compatible repair fabric, a second layer of biologically compatible repair fabric attached to the first layer, and a plurality of tissue grips protruding from a first surface of the first layer. The plurality of grips are independent of the first layer and secured to the first layer with the second layer.

According to another aspect, an implantable prosthesis comprises a first layer of biologically compatible repair fabric and a plurality of tissue grips protruding from a first

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surface of the first layer. The plurality of grips are configured to penetrate and grip tissue. Each tissue grip includes a grip body protruding from the first layer and a grip head located at an end of the grip body spaced away from the first surface of the first layer. The grip head includes a first pair of barbs located along a first axis oriented in a first radial direction and a second pair of barbs located along a second axis oriented in a second radial direction that is different from the first radial direction.

According to another aspect, an implantable prosthesis comprises a first layer of biologically compatible repair fabric and a plurality of tissue grips protruding from a first surface of the first layer. The plurality of grips are configured to penetrate and grip tissue. Each tissue grip includes a grip body protruding from the first layer and a grip head located at an end of the grip body spaced away from the first surface of the first layer. The grip head includes a plurality of primary barbs and a plurality of secondary barbs with the primary barbs being different from the secondary barbs. The primary barbs are configured to minimize entanglement of the grip head with the repair fabric. The primary barbs may be positioned to shield the secondary barbs from entanglement.

According to another aspect, a method is provided for fabricating an implantable prosthesis. The method comprises acts of: (a) providing a first layer of biologically compatible repair fabric including a first surface and a second surface opposite the first surface, and (b) attaching a plurality of tissue grips to the first layer of repair fabric. The plurality of grips are fabricated independent of the first layer. Each tissue grip includes a grip base and a grip body extending from the grip base. The grips are attached to the first layer by passing the grip body through the first layer to protrude beyond the first surface and positioning the grip base adjacent the second surface.

The foregoing is a non-limiting summary of the invention, which is defined by the attached claims. Other aspects, embodiments and/or features will become apparent from the following description.

Various embodiments of the present invention may provide certain advantages and may overcome certain drawbacks of prior prostheses. Embodiments of the invention may not share the same advantages, and those that do may not share them under all circumstances.

BRIEF DESCRIPTION OF DRAWINGS

Aspects of the invention are described below, by way of example, with reference to the accompanying drawings in which like numerals reference like elements, and wherein:

FIG. 1 is an end perspective view of an implantable prosthesis according to one embodiment;

FIG. 2 is a side perspective view of the implantable prosthesis of FIG. 1;

FIG. 3 is a top view of the implantable prosthesis of FIG. 1;

FIG. 4 is a bottom view of the implantable prosthesis of FIG. 1;

FIG. 5 is a sectional view taken along section line 5-5 of FIG. 4 illustrating a tissue grip being attached to the prosthesis according to one embodiment;

FIG. 6 is a sectional view taken along section line 5-5 of FIG. 4 illustrating a tissue grip being attached to the prosthesis according to another embodiment;

FIG. 7 is a perspective view of a tissue grip of FIGS. 5-6 according to one illustrative embodiment;

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FIG. 8 is a perspective view of a tissue grip according to another illustrative embodiment;

FIG. 9 is a side elevational view of the tissue grip of FIG. 8;

FIG. 10 is a side elevational view of the tissue grip of FIG. 7;

FIG. 11 is a side elevational view of the tissue grip of FIG. 10 rotated 90°;

FIG. 12 is a top view of a grip assembly according to one illustrative embodiment;

FIG. 13 is a perspective view of a tissue grip according to another illustrative embodiment;

FIG. 14 is an enlarged side view of the grip head of FIG. 13 according to one embodiment;

FIG. 15 is a side elevational view of a tissue grip according to another illustrative embodiment;

FIG. 16 is an enlarged side view of the grip head of FIG. 15 according to one embodiment;

FIG. 17 is an enlarged bottom view of the grip head of FIGS. 15-16 according to one embodiment;

FIG. 18 is a perspective view of a tissue grip with a V-shaped configuration according to another illustrative embodiment;

FIG. 19 is a perspective view of a tissue grip with a crescent shaped configuration according to another illustrative embodiment;

FIG. 20 is a perspective view of a tissue grip with a claw-like configuration according to another illustrative embodiment;

FIG. 21 is a side elevational view of a tissue grip with an arrowhead configuration according to another illustrative embodiment;

FIG. 22 is a top view of an implantable prosthesis with a grip arrangement according to one embodiment;

FIG. 23 is a top view of an implantable prosthesis with a grip arrangement according to one embodiment;

FIG. 24 is a top view of an implantable prosthesis with a grip arrangement according to one embodiment;

FIG. 25 is an illustration of a grip arrangement according to another embodiment;

FIG. 26 is an illustration of a grip arrangement according to another embodiment;

FIG. 27 is an illustration of a grip arrangement according to another embodiment;

FIG. 28 is an illustration of a grip arrangement according to another embodiment;

FIG. 29 is an illustration of a grip arrangement according to another embodiment;

FIG. 30 is an illustration of a grip arrangement according to another embodiment;

FIG. 31 is an illustration of a grip arrangement according to another embodiment; and

FIG. 32 is an illustration of a grip arrangement according to another embodiment.

DETAILED DESCRIPTION

It should be understood that aspects of the invention are described herein with reference to the figures, which show illustrative embodiments in accordance with aspects of the invention. The illustrative embodiments described herein are not necessarily intended to show all aspects of the invention, but rather are used to describe a few illustrative embodiments. Thus, aspects of the invention are not intended to be construed narrowly in view of the illustrative embodiments. It should be appreciated, then, that the various concepts and embodiments discussed herein may be implemented in any

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of numerous ways, as the disclosed concepts and embodiments are not limited to any particular manner of implementation. In addition, it should be understood that aspects of the invention may be used alone or in any suitable combination with other aspects of the invention.

The invention is directed to an implantable prosthesis for mending an anatomical defect, and is particularly suitable for mending defects in, and weaknesses of, soft tissue and muscle walls or other anatomical regions. The phrase “mending a defect” includes acts of repairing, augmenting, and/or reconstructing a defect and/or a potential defect. For ease of understanding, and without limiting the scope of the invention, the prosthesis is described below particularly in connection with mending a groin defect including, but not limited to, one or more of an indirect inguinal hernia, a direct inguinal hernia, a femoral hernia and/or other weakness or rupture of the groin anatomy. It should be understood, however, that the prosthesis is not so limited and may be employed in other anatomical procedures, as should be apparent to one of skill in the art. For example, and without limitation, the prosthesis may be employed for ventral hernias, chest or abdominal wall reconstruction, or large defects, such as those that may occur in obese patients. The prosthesis may include one or more features, each independently or in combination, contributing to such attributes.

The invention is more particularly directed to a prosthesis which includes a repair fabric having a body portion that is configured to cover or extend across the defect opening or weakness when the body portion is placed against the defect. The prosthesis may be in the form of a patch, although the prosthesis may employ other configurations as should be apparent to one of skill in the art. The patch may have a planar or non-planar configuration suitable for a particular procedure employed for mending a defect.

The prosthesis may be configured with a self-grip arrangement having features that help maintain the position of the prosthesis relative to the defect. The self-grip arrangement may reduce, if not eliminate, separation, sliding, twisting, folding and/or other movement, as may be desired, between the prosthesis and adjacent tissue. Such an arrangement may also reduce, if not eliminate, the need for a surgeon to suture, staple, tack, or otherwise provisionally anchor the prosthesis in place pending tissue integration.

The prosthesis may include a plurality of grips protruding from the body portion. More particularly, the grips may protrude from a surface of the body portion that is configured to engage adjacent tissue. The grips may be configured to penetrate and grip the tissue when the prosthesis is placed and/or pressed against it. In this manner, the grips may be configured to protrude a defined distance from the surface of the body portion to penetrate a depth of tissue sufficient to provide the desired amount of grip.

The grips may be arranged on the body portion in any suitable configuration to provide a desired amount of grip as should be apparent to one of skill in the art. For example, and without limitation, the grips may be distributed across the body portion in a uniform, non-uniform or random array, and/or any suitable combination of arrays. The grips may be distributed across the entire body portion or located at one or more select regions of the body portion. For example, and without limitation, the grips may be located at one or more select regions adjacent one or more segments of the outer periphery of the body portion, and/or one or more select regions located within the inner region of the body portion inwardly away from the outer periphery. Each select region may include one or more grips arranged in any suitable pattern within the region. One or more of the select regions

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may employ the same or different arrangements of grips relative to one or more other select regions of grips.

According to one aspect, the grips may be fabricated independent of and mounted to the body portion of the prosthesis. In this manner, the grips may be formed from a material that is different from the body portion. For example, and without limitation, the grips may be formed of a bioabsorbable material, while the body portion may be formed of a non-absorbable material. Such an arrangement may provide the prosthesis with temporary grip properties during the period of tissue integration, while reducing the amount of foreign material that remains present in a patient's body and maintaining long-term strength of the prosthesis.

Independent fabrication of the grips may also provide flexibility for configuring the prosthesis. For example, and without limitation, the prosthesis may include grips having the same or different grip configurations and/or arrangements depending on a particular application of the prosthesis. For example, and without limitation, the prosthesis may include grips having the same shape, but mounted in different orientations relative to each other on the body portion. The prosthesis may include grips with one or more different shapes in one or more regions of the body portion. In this manner, the prosthesis may be provided with various grip characteristics based on the particular orientations and/or shapes of the grips individually and as a whole.

The grips may be directly or indirectly mounted to the body portion of the prosthesis. Each grip may include a base and a grip body extending from the base. The base may be mounted to the body portion with the grip body protruding from the body portion and configured to penetrate and grip tissue.

In one aspect, the grips may be molded or welded directly to the body portion. For example, and without limitation, each grip may be insert molded to prefabricated mesh fabric placed within a mold that receives grip material and forms the desired shape of the grip directly onto the body portion.

In one aspect, each grip may be mechanically mounted to the body portion. For example, and without limitation, each grip may be mounted using a mechanical component or arrangement that is attached to the body portion. For example, and without limitation, the body portion may include a first layer of repair fabric and a second layer of repair fabric attached to the first layer to secure the grips to the first layer. The base of each grip may be retained between the first and second layers of repair fabric. Alternatively, each grip may be mounted directly to the second layer which is subsequently attached to the first layer. Such indirect mounting arrangements may reduce the likelihood of potential degradation of the strength and/or tissue infiltration characteristics of the first layer. In another non-limiting example, the base of each grip may be configured to mechanically connect with the repair fabric of the body portion.

The grip may include a grip head located at the end of the grip body opposite the base that is configured for insertion into tissue while providing a sufficient amount of tissue grip to reduce, if not eliminate, inadvertent release from the tissue. The grip head may also be configured to minimize, if not eliminate, potential entanglement with the prosthesis, such as a mesh repair fabric, while providing desired tissue grip. For example, and without limitation, the grip head may include one or more barbs of any suitable configuration as should be apparent to one of skill in the art. The grip head may include barbs having different configuration relative to each other. For example, and without limitation, a first barb

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configuration may be suitable for tissue grip while minimizing entanglement and a second barb configuration may be suitable for just gripping tissue. The grip head configurations may include, but are not be limited to, an arrowhead shape, a crescent shape, multiple prongs arranged in a V-shape, and a claw shape. Other grip head configurations are also contemplated.

The prosthesis may be used for mending soft tissue and muscle wall defects using various surgical techniques, including open, laparoscopic, hybrid (e.g., Kugel procedure), and robotic techniques. During open procedures, the prosthesis may be placed through a relatively large incision made in the abdominal wall and layers of tissue and then the defect is filled or covered with the repair fabric. During laparoscopic and hybrid procedures, the prosthesis may be collapsed, such as by rolling or folding, into a reduced configuration for entry into a subject, either directly through a comparatively smaller incision or through a slender laparoscopic cannula that is placed through the incision. The prosthesis may have particular application with robotic procedures in which placement of the prosthesis is achieved using surgical robotic tools which may involve passage of the prosthesis through a relatively small cannula (e.g., 8 mm) as compared to a cannula (e.g., 10-12 mm) typically employed for more conventional laparoscopic techniques.

FIGS. 1-4 illustrate one embodiment of a prosthesis for mending tissue and muscle wall defects, such as a hernia defect. The prosthesis includes a repair fabric of implantable, biologically compatible material. In one embodiment, the repair fabric may comprise a mesh fabric that is relatively flexible, thin and light weight and meets the performance and physical characteristics for mending soft tissue and muscle wall defects.

The prosthesis **20** may include a body portion **22** configured with a size and/or shape suitable to cover or extend across the defect opening or weakness when the body portion is placed against the defect. The prosthesis may also include a plurality of grips **24** protruding from the body portion to provide a self-grip arrangement for maintaining the position of the prosthesis relative to the defect. The grips **24** are configured to protrude from a surface **25** of the body portion for engaging adjacent tissue.

The grips **24** may be configured to penetrate and grip the tissue when the prosthesis is placed and/or pressed against it. In this manner, the grips may be configured to protrude a defined distance from the surface of the body portion to penetrate a depth of tissue sufficient to provide the desired amount of grip. In one embodiment, the grips **24** may be configured to protrude 0.05 inches to 0.15 inches from the surface. However, the grips may be configured to protrude other distances sufficient to provide a desired amount of grip as should be apparent to one of skill in the art.

In one illustrative embodiment, the grips **24** may be located at one or more grip regions **26** of the body portion. Such an arrangement may be suitable for placing grips in selected regions of the body portion to accommodate a particular anatomical region. For example, it may be desirable to avoid providing grips on regions of the body portion that may potentially contact vessels, nerves or other portions of the anatomy at the defect site as should be apparent to one of skill in the art.

As shown in FIGS. 1-4, the grip regions **26** may be located adjacent one or more segments of the outer periphery **28** of the body portion **22** and extend inwardly toward the inner region of the body portion. Each grip region **26** may include a plurality of grips **24** arranged in any suitable pattern within the region, for example, to facilitate rolling

and/or folding of the prosthesis to the site of the soft tissue repair. In one illustrative embodiment, the prosthesis **20** may be provided with three grip regions **26** including grips **24** arranged in multiple rows and/or columns with the number of grips in each row and/or column decreasing in a direction away from the outer periphery. As shown, each grip region **26** may include ten grips **24** arranged in four rows and/or columns of decreasing grips. In this manner, the grips may be viewed as having a generally triangular arrangement.

It is to be appreciated that any suitable grip arrangement may be provided on the prosthesis to provide a desired amount of grip as should be apparent to one of skill in the art. For example, and without limitation, a single row of grips may be located along one or more select segments of the outer periphery. In other embodiments, one or more of the select regions may employ the same or different arrangements of grips relative to one or more other select regions of grips. The grips may be arranged in a uniform, non-uniform or random array, and/or any suitable combination of arrays. Rather than limited to one or more select grip regions, the grips may be distributed across the entire body portion.

The grips may be fabricated independent of and mounted to the body portion of the prosthesis. Independent fabrication of the grips may provide flexibility for configuring the prosthesis.

In one embodiment, the grips may be formed of a bioabsorbable material, while the body portion may be formed of a non-absorbable material. Such an arrangement may provide the prosthesis with temporary grip properties during the period of tissue integration, while reducing the amount of foreign material that remains present in a patient's body and maintaining long-term strength of the prosthesis.

In other embodiments, the prosthesis may include grips having the same or different grip configurations and/or arrangements depending on a particular application of the prosthesis. For example, and without limitation, the prosthesis may include grips having the same shape, but mounted in different orientations relative to each other on the body portion. The prosthesis may include grips with one or more different shapes in one or more regions of the body portion. In this manner, the prosthesis may be provided with various grip characteristics based on the particular orientations and/or shapes of the grips individually and as a whole.

In one illustrative embodiment shown in FIG. 5, each grip **24** may include a base **30** and a grip body **32** extending from the grip base. The grip base **30** may be mounted to the body portion **22** of the prosthesis with the grip body **32** protruding from the body portion to penetrate and grip tissue. The body portion **22** of the prosthesis may include a first layer **34** of repair fabric configured to extend across and cover the defect and a second layer **36** of repair fabric attached to the first layer to secure each grip to the first layer. As shown, the base **30** of each grip may be retained between the first and second layers **34**, **36** of repair fabric with the second layer **36** acting as a backing layer. The grip body **32** may extend through a pore **38** of the first layer **34** of fabric, although the prosthesis is not so limited.

In one embodiment, each backing layer **36** may be configured to correspond with the shape of each grip region **26** of the prosthesis. However, the shape of the backing layer is not so limited and may have any suitable configuration apparent to one of skill in the art.

Each backing layer **36** may be attached to the first layer **34** using any suitable fastening arrangement as should be apparent to one of skill in the art. In one embodiment, the backing layers **36** may be attached to the first layer **34** using stitches **40** extending about the outer periphery **42** of the

backing layer. Additional stitches may be placed between rows and/or columns of the grips to reduce billowing of the backing layer and maintain the grips in position between the layers. It is to be appreciated that other attachment techniques may be employed including, but not limited to, bonding and ultrasonic welding.

Rather than securing the grips between layers of repair fabric, it may be desirable to directly mount the grips to a second layer for subsequent attachment to the first layer. Such an arrangement may be desirable to allow one or more grip regions of any desired configuration be trimmed from a relatively large layer of fabric having pre-attached grips. Such an indirect mounting arrangement may also reduce the incidence of potential degradation of the strength and/or tissue infiltration characteristics of the first layer.

In one illustrative embodiment shown in FIG. 6, each grip **24** may be mounted directly to the second layer **36** of fabric which is subsequently attached to the first layer **34** of fabric. As shown, the grip base **30** may be secured to the second layer **36** with the grip body **32** extending through the second layer. The grip base may be attached to the second layer using any suitable technique including, but not limited to, bonding or ultrasonic welding **44**. The second layer may be attached to the first layer using any suitable fastening technique as described above.

It is to be understood that other techniques may be employed to attach tissue grips to the body portion of the prosthesis. In one embodiment, the grips may be molded or welded directly to the body portion. For example, and without limitation, each grip may be insert molded to prefabricated mesh fabric placed within a mold that receives grip material and forms the desired shape of the grip directly onto the mesh fabric. The mesh fabric may be used as the first layer **34** of the body portion or as the second layer **36** of the body portion that is attached to the first layer. In another non-limiting example, the base **30** of each grip **24** may be configured to mechanically connect with the repair fabric of the body portion.

As indicated above, the prosthesis may include grips **24** having the same or different grip configurations and/or arrangements depending on a particular application of the prosthesis.

In one embodiment illustrated in FIGS. 5-7, the grip **24** may include a grip head **50** located at the end of the grip body **32** opposite the base **30**. The grip head **50** may be configured for insertion into tissue while providing a sufficient amount of tissue grip to reduce, if not eliminate, inadvertent release from the tissue. The grip head may include one or more barbs configured to penetrate tissue and resist withdrawal of the grip from the tissue. In one embodiment, the grip head **50** may include four barbs **52a**, **52b** located at 90° positions about the end of the grip body. In this manner, the grip head may employ a cruciform configuration with a first pair of barbs **52a** located along a first axis Y and a second pair of barbs **52b** located along a second axis X. Such an arrangement may provide omnidirectional sliding resistance. However, the grip may include any number and location of barbs as should be apparent to one of skill in the art.

For some applications, it may be desirable to vary the location of the barbs along the length L_1 of the grip body. For example, it may be desirable to grip the tissue at different depths of penetration. In one embodiment, the grip head may include a first pair of barbs **52a** located at a first distance L_2 along the grip body and a second pair of barbs **52b** located at a second distance L_3 along the grip body which is different from the first distance. As should be appreciated, any

number of barbs may be located at different locations along the barb body as should be apparent to one of skill in the art.

To facilitate locating and/or holding a tissue grip in position on the body portion, it may be desirable to provide the grip with a self-locating feature. In one embodiment illustrated in FIGS. 5-7, the grip may include a grip locator 54 configured for insertion into a pore 38 or other opening in the first or second layer of repair fabric. The size and/or shape of the grip locator may be selected to correspond to the pores of the repair fabric. As illustrated, the grip locator 54 may be configured as a boss protruding from the grip base 30 with the grip body extending from the locator. The grip locator 54 may protrude from a central region of the grip base, although other arrangements are contemplated.

In one embodiment shown in FIG. 7, the grip locator 54 may have a cylindrical shape, although other shapes are contemplated. For example, in one embodiment illustrated in FIGS. 8-9, the grip locator 54 may have a conical or frusto-conical shape. As shown, the grip locator 54 may be tapered to decrease in size in a direction from the grip base 30 toward the grip body 32. Such an arrangement may facilitate insertion of the tissue grip into a pore 38 or other opening of the repair fabric. It should be understood, however, that the grip locator, if provided, may employ any suitable configuration as should be apparent to one of skill in the art.

In one illustrative embodiment shown in FIGS. 10-11, the grip base 30 may have a diameter D_1 of about 0.125 inches with a thickness T_1 of about 0.01 inches. The grip locator may have a diameter D_2 of about 0.05 inches with a thickness T_2 of about 0.01 inches. The grip body may have a length L_1 of about 0.08 inches extending from the surface of the grip base and a diameter D_3 of about 0.02 inches to about 0.035 inches. The grip head may include a first pair of barbs 52a with a grip surface 56a located a distance L_2 of about 0.06 inches from the grip base and a second pair of barbs 52b with a grip surface 56b located a distance L_3 of about 0.047 to about 0.05 inches from the grip base. The first pair of barbs 52a may have an outer diameter D_4 of about 0.04 inches and the second pair of barbs 52b may have an outer diameter D_5 of about 0.044 inches. The first pair of barbs 52a may have a width W_1 of about 0.006 inches and the second pair of barbs 52b may have a width W_2 about 0.01 inches.

In the embodiments described above, each grip may be fabricated as a single grip that is attached to the body portion of the prosthesis. For some applications, it may be desirable to fabricate and provide multiple interconnected grips for attachment to the body portion. For example, a grip arrangement including multiple grips may facilitate handling and assembly of the grips to the body portion in contrast to the smaller size of a single grip.

In one embodiment illustrated in FIG. 12, a grip assembly 60 may include two (as shown) or more grips 24 connected together with an elongated coupling 62 extending therebetween. In one embodiment, the grip assembly may be fabricated as a unitary component. For example, the grip assembly may be a molded component fabricated from the same material. However, the grip assembly may be fabricated using any suitable process and may include multiple materials depending on the particular characteristics desired for the grips.

The coupling may be configured to provide a desired amount of support for handling the grips while also having a sufficient amount of flexibility to minimize any potential increase in the stiffness of the body portion. In one embodiment, the coupling 62 may have a thickness that corresponds

to the thickness of the grip base 30 of the grips. The coupling 62 may be configured with a width W_3 of about 0.04 inches. The grips 24 may be configured with a center-to-center spacing L_4 of about 0.38 inches. It is to be understood that any suitable coupling arrangement and grip spacing may be utilized as should be apparent to one of skill in the art.

For some applications, it may be desirable to employ a grip head 50 with a cruciform shape including barbs 52a, 52b having the same size and a grip surface 56 located the same distance from the grip base 30. FIGS. 13-14 illustrate one embodiment of such an arrangement. The grip head 50 may be utilized for a single grip, as illustrated in FIG. 13, or on each grip of a grip assembly, such as the grip assembly described above.

For some applications, it may be desirable to employ a grip head configured to reduce, if not eliminate, potential mesh entanglement during handling and/or delivery of the prosthesis. For example, and without limitation, it may be desirable to fold, roll, or otherwise collapse the prosthesis to facilitate delivery of the prosthesis during some surgical procedures. When collapsed, at least some of the grips may be placed into engagement with other portions of the prosthesis which could potentially lead to a grip becoming entangled with the prosthesis, particularly when the prosthesis is fabricated from a mesh fabric. Such entanglement may make it more difficult and require additional time for a surgeon to open the collapsed prosthesis after delivery to the surgical site.

In one embodiment illustrated in FIGS. 15-17, the prosthesis may include grips 24 having a grip head 50 configured to grip tissue while also minimizing potential entanglement with the prosthesis. The grip head may include one or more primary barbs 52a configured to both grip tissue and minimize, if not eliminate, potential entanglement with the prosthesis. The grip head may also include one or more secondary barbs 52b configured solely to grip tissue. In this manner, the secondary barbs may enhance the grip of the barbs by providing additional traction while the primary barbs minimize potential entanglement.

The primary barbs 52a may be configured and arranged to shield the secondary barbs 52b from entanglement with the prosthesis while allowing the secondary barbs to grip tissue. In one embodiment, the secondary barbs 52b may be smaller in size as compared to the primary barbs 52a. As shown in FIG. 17, the grip head 50 may include four primary barbs 52a located along axes X, Y positioned 90° apart and four secondary barbs 52b located along axes X', Y' positioned 90° apart with the primary barbs being offset from the secondary barbs by about 45°. In this manner, each secondary barb 52b may be located between a pair of primary barbs 52a to minimize, if not prevent, engagement between the secondary barbs and the prosthesis, such as the filaments of a mesh prosthesis. However, it is to be appreciated that the grip head 50 may employ any number of primary barbs and/or secondary barbs positioned in any suitable arrangement as should be apparent to one of skill in the art.

The primary grip 52a may include a first grip surface 56a to grip tissue when placed against tissue while allowing the grip head 50 to release from the prosthetic material, such as mesh fabric, when placed against the prosthesis. In one embodiment, the first grip surface 56a may be configured with a ramp-like structure from which filaments of the prosthesis may be slid off and away from the grip head as the grip is being pulled away from the prosthesis. As shown, the first grip surface may be angled to slope inwardly from the outer tip 58a of the primary barb toward the grip body 32 and in a direction toward the grip base 30. In this manner, the

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first grip surface **56a** may be oriented at an angle A so that filaments of the prosthesis located along the length of the grip body may be guided along the grip surface and off the grip head as the grip is pulled from the prosthesis.

In one embodiment, the first grip surface **56a** may be oriented at an angle A of 30° relative to a plane P perpendicular to the longitudinal axis Z of the grip. However, other angles suitable for tissue grip and grip head release from the prosthesis may be employed as should be apparent to one of skill in the art. In this manner, varying the first grip surface angle may adjust the amount of tissue grip and grip head release. For example, a relatively large grip surface angle may enhance grip head release while reducing the amount of tissue grip, while a relatively small grip surface angle may reduce grip head release while increasing the amount of tissue grip.

The secondary grip **52b** may include a second grip surface **56b** configured to grip tissue when placed against tissue. In contrast to the angled configuration of the first grip surface **56a**, the second grip surface may have a relatively flat configuration. In one embodiment, the second grip surface **56b** may be oriented perpendicular to the longitudinal axis Z of the grip to provide a relatively higher degree of tissue grip as compared to the first grip surface **56a**. If desired, other angles suitable for tissue grip may be employed as should be apparent to one of skill in the art. In this manner, varying the angle of the secondary grip surface may adjust the amount of tissue grip desired for the grip head.

In one embodiment as shown, the first grip surface **56a** may be angled from the plane P of the second grip surface **56b**. In this manner, the outer tips **58a** of the primary grips **52a** and the outer tips **58b** of the secondary grips **52b** may be coplanar. However, it is to be appreciated that the outer tip **58a** of the primary grip **52a** may be located in a plane offset from the tip **58b** of the secondary grip **52b**.

As indicated above, the primary grips **52a** may be configured to shield the secondary grips **52b** in a manner to minimize entanglement of the secondary grips with the prosthesis. In one embodiment shown in FIGS. 15-17, the first grip surface **56a** may be located between the second grip surface **56b** and the grip base **30**. Additionally, the tips **58a** of the primary grips **52a** may extend in an outward radial direction from the grip body **32** a distance which is greater than the tips **58b** of the secondary grips **52b**. In this manner, features of the prosthesis, such as filaments of a mesh fabric, may be guided by the first grip surface in the outward radial direction away from the grip body and beyond the tip **58b** of the secondary grip so that the filaments are guided around the secondary grips as the grip is pulled from the prosthesis. In one embodiment, the outer tips **58a** of the first grips may be positioned along a circular diameter D_6 of 0.042 in and the outer tips **58b** of the secondary tips may be positioned along a circular diameter D_7 of 0.031 in. However, it is to be appreciated that other arrangements of the outer tips of the barbs of the grip head may be employed as should be appreciated by one of skill in the art.

The grip head **50** may be utilized for a single grip, as illustrated in FIG. 15, or on each grip of a grip assembly, such as the grip assembly described above.

As indicated above, the tissue grip may employ various configurations to provide a desired amount of grip for holding the prosthesis in place against tissue. The grip described above may be considered as having a cruciform configuration. Examples of other non-limiting grip configurations are described below in connection with FIGS. 18-21.

FIG. 18 illustrates an embodiment of a grip having a multi-prong configuration. As shown, the grip body **32** may

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include a pair of prongs **70** extending from the grip base **30**. The prongs **70** may be arranged in a V-shaped configuration for penetration into tissue.

FIG. 19 illustrates an embodiment of a grip having a crescent configuration. As shown, the grip body **32** may have a crescent shape extending from the grip base **30**. The grip body **32** decreases in size in a direction from the base **30** toward its tip **72** for penetration into tissue.

FIG. 20 illustrates an embodiment of a grip having a claw-like configuration. As shown, the grip body **32** employ a C-shaped configuration with a pair of tips **74** for penetrating tissue. As shown, the grip body **32** may be oriented relative to the grip base **30** with the tips extending in a direction away from the grip base.

FIG. 21 illustrates an embodiment of a grip having an arrowhead configuration. As shown, the grip body **32** extends from the grip base **30**. The grip body **32** include an arrowhead with a pair of oppositely extending barbs for penetrating and gripping tissue.

As illustrated in FIGS. 1-4, the body portion **22** may be a preformed, non-planar patch with a 3-dimensional curved shape. In one embodiment, the body portion **22** may have a shape corresponding to the 3DMAX Light Mesh or 3DMAX Mesh, available from C. R. Bard, and described in one or more of U.S. Pat. Nos. 6,723,133, 6,740,122 and 6,740,122. In this manner, the prosthesis may be particularly suited for fitting and mending defects to the inguinal anatomy. However, it is to be understood that the prosthesis may employ other configurations as should be apparent to one of skill in the art. For example, and without limitation, the patch may have planar or other non-planar configurations suitable for a particular procedure employed for mending a defect. Moreover, the prosthesis may be provided as a planar sheet of self-gripping repair fabric that may be selectively trimmed by a surgeon to any desired size and shape for the particular procedure.

As indicated above, the prosthesis may employ any arrangement of grips to provide a desired amount of grip and traction for a particular application.

FIG. 22 illustrates an embodiment of a 3D curved prosthesis **20** including four separate grip regions **26**, as compared to three grip regions as shown in FIGS. 1-4, located in proximity to the outer periphery **28** of the body portion **22**. As shown, each group or grip region **26** may include multiple rows and/or columns of grips **24**.

FIG. 23 illustrates an embodiment of a 3D curved prosthesis **20** including three separate grip regions **26** located in proximity to the outer periphery **28** of the body portion **22**. A pair of grip regions may be located on upper medial and lateral portions of the prosthesis, and a grip region may be located on a lower medial portion of the prosthesis. As shown, each group or grip region **26** may include multiple rows and/or columns of grips **24**. Each grip region on the upper medial and lateral portions may have a semi-annular periphery. The grip region on the lower medial portion may have a triangular periphery.

FIG. 24 illustrates an embodiment of a 3D curved prosthesis **20** including separate groups of grips **24** located along the outer periphery **28** of the body portion **22**. As shown, each group or grip region **26** may include a single row of grips **24**.

FIG. 25 illustrates an embodiment of a layer of repair fabric including grips **24** uniformly distributed across the fabric. The fabric may be preformed into a configuration for use as a prosthesis or may be selectively trimmed into a desirable configuration for use as a prosthesis or a portion of a prosthesis, such as a second layer described above. As

shown, the grips may be mounted to the fabric in at least two different orientations to provide a desired grip or traction properties for the prosthesis. For example, and without limitation, the prosthesis may include grips, such as the grips illustrated in FIGS. 18-21, which individually provide bidirectional traction. Mounting the grips in alternating 90° orientations may provide omnidirectional traction for the prosthesis. The particular spacing and orientation may be selected to provide the desired grip and/or traction.

The repair fabric may employ a knit construction that provides openings or pores to allow tissue infiltration to incorporate the prosthesis. The repair fabric may also have sufficient flexibility to promote an easy reduction in size for entry into the subject. In this manner, the flexible fabric may be collapsed into a slender configuration, such as a roll, which can be supported in, and advanced through, a narrow laparoscopic cannula for use in laparoscopic or robotic procedures.

In one embodiment, the prosthesis 20 may be formed from one or more layers of knitted mesh fabric. When implanted, the mesh promotes rapid tissue or muscle ingrowth into and around the mesh structure. Examples of surgical materials which may be utilized for the layers and are suitable for tissue or muscle reinforcement and defect correction include, but are limited to, BARD Mesh (available from C. R. Bard, Inc.), BARD Soft Mesh (available from C. R. Bard, Inc.), SOFT TISSUE PATCH (microporous ePTFE—available from W. L. Gore & Associates, Inc.); SURGIPRO (available from US Surgical, Inc.); TRELEX (available from Meadox Medical); PROLENE and MERSILENE (available from Ethicon, Inc.); and other mesh materials (e.g., available from Atrium Medical Corporation). Absorbable or resorbable materials, including PHASIX Mesh (available from C. R. Bard, Inc.), polyglactin (VICRYL—available from Ethicon, Inc.) and polyglycolic acid (DEXON—available from US Surgical, Inc.), may be suitable for applications involving temporary correction of tissue or muscle defects. Collagen materials such as COOK SURGISIS, available from Cook Biomedical, Inc. may also be used. It also is contemplated that the mesh fabric may be formed from multifilament yarns and that any suitable method, such as knitting, weaving, braiding, molding and the like, may be employed to form the mesh material.

In one embodiment, the grips may be formed of a resorbable material. In this manner, the grips provide a desired level of tissue grip for initial placement and positioning of the prosthesis. The grips eventually become resorbed by the body at which time the prosthesis has become sufficiently integrated with the tissue. Examples of resorbable materials which may be utilized for the grips include, but are not limited to, PLG (L-lactide/glycolide) or PLA (polylactic acid) polymers. For some applications, it may be desirable to utilize non-resorbable grips. An example of non-resorbable material which may be utilized for the grips includes, but is not limited to, polypropylene. It is to be appreciated that other suitable biocompatible materials, resorbable or non-resorbable, may be used for the grips as should be apparent to one of skill in the art.

In one embodiment, the grips may be fabricated using an injection molding process. However, any suitable manufacturing process may be used to fabricate the grips as should be apparent to one of skill in the art. For example, and without limitation, the grips may be fabricated using 3D printing and machining processes.

Examples

The following examples are illustrative only and are not intended to limit the scope of the present invention.

Traction force was evaluated for various grip arrangements and compared to a known mesh fabric. Testing methodology is described below with results provided in Table 1. The average results are reported from testing ten samples of each configuration.

A fresh, square coupon of porcine tissue, kept at room temperature, was prepared and placed in a test fixture. The porcine tissue coupon measured 2.7 inches×3.5 inches with a minimum thickness of 0.25 inches. An approximately 1 inch×1 inch test sample (unless indicated otherwise below) of a self-gripping mesh configuration was placed, barb side down, on top of and centered on the porcine tissue sample. A free weight was set on top of the mesh sample for about 3 seconds and then removed. The free weight was approximately 1 lb to 3 lb. The mesh sample was allowed to rest/settle on the porcine tissue sample, at room temperature, undisturbed for 1 minute before initiating the test.

Using a force gauge, such as a Mark-10 tensiometer gauge, with a gripping attachment to secure a portion of the mesh sample, a traction force was applied to the mesh in a single direction along the plane defined by the contact area between the mesh sample and the porcine tissue sample. The mesh was oriented so that the grips were pulled in the direction of the barbs or other grip features. The procedure was repeated for ten mesh samples of the same configuration.

Test samples were provided with different grip configurations and arrangements of grips. The grip arrangements of the test samples are illustrated in FIGS. 26-31. The direction of applied force is shown as arrow F.

FIG. 26 illustrates a grip arrangement including grips located every other space and every other row. This arrangement was tested for grip configurations illustrated in FIGS. 18-21.

FIG. 27 illustrates a grip arrangement including grips located every other space and every row. This arrangement was tested for grip configurations illustrated in FIGS. 18-21.

FIG. 28 illustrates a grip arrangement including grips positioned in a 5×5 square. This arrangement was tested for grip configurations illustrated in FIGS. 18-21.

FIG. 29 illustrates a grip arrangement including grips positioned in a 4×4 square. This arrangement was tested for the grip configuration illustrated in FIG. 21.

FIG. 30 illustrates a grip arrangement including grips located in two rows with four grips provided in one row and three grips provided in the other row. This arrangement was tested for the grip configuration illustrated in FIG. 21.

FIG. 31 illustrates a grip arrangement including grips located in two rows with eight grips provided in each row. This arrangement was tested for grip configurations illustrated in FIGS. 18-21.

FIG. 32 illustrates a grip arrangement including seven grip arrays located in three rows with three arrays provided in a center row flanked by rows with two arrays. The grip arrays were located within a test sample area of approximately 1.7 inches×0.8 inches. Each array includes three grips for a total of twenty-one grips with nine grips located along the central row and six grips located along each row flanking the center row. This arrangement was tested for the grip configuration illustrated in FIGS. 15-17. This arrangement provided an average traction force of 0.68 lbf.

For ease of presenting the test results for the other arrangements, the grip configurations are referred to as Grip A (FIG. 18), Grip B (FIG. 19), Grip C (FIG. 20) and Grip D (FIG. 21). Similarly, the grip arrangements are referred to as Arrangement 1 (FIG. 26), Arrangement 2 (FIG. 27),

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Arrangement 3 (FIG. 28), Arrangement 4 (FIG. 29), Arrangement 5 (FIG. 30), and Arrangement 6 (FIG. 31).

TABLE 1

Test Sample	Average Traction Force (psi)
Covidien ProGrip Mesh	0.275
Grip A, Arrangement 1	0.285
Grip A, Arrangement 2	0.335
Grip B, Arrangement 1	0.3
Grip B, Arrangement 2	0.26
Grip C, Arrangement 1	0.285
Grip C, Arrangement 2	0.39
Grip A, Arrangement 3	0.335
Grip B, Arrangement 3	0.21
Grip C, Arrangement 3	0.315
Grip D, Arrangement 4	0.49
Grip D, Arrangement 5	0.32
Grip A, Arrangement 6	0.335
Grip B, Arrangement 6	0.365
Grip C, Arrangement 6	0.415

For purposes of this patent application and any patent issuing thereon, the indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified.

The use of “including,” “comprising,” “having,” “containing,” “involving,” and/or variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

The foregoing description of various embodiments are intended merely to be illustrative thereof and that other embodiments, modifications, and equivalents are within the scope of the invention recited in the claims appended hereto.

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What is claimed is:

1. A method of fabricating an implantable prosthesis for implantation in a person, the method comprising acts of:

(a) providing a first layer of biologically compatible repair fabric including a first surface and a second surface opposite the first surface, the first layer having a 3-dimensional curved configuration adapted to fit an inguino-femoral anatomy of a person;

(b) attaching a plurality of preformed tissue grips to the first layer of repair fabric to form a self-gripping prosthesis prior to implantation, the plurality of preformed tissue grips being fabricated independent of the first layer, each preformed tissue grip including a grip base and a grip body extending from the grip base, each of the plurality of preformed tissue grips being attached to the first layer by passing the grip body through the first layer to protrude beyond the first surface and positioning the grip base adjacent the second surface; and

(c) attaching a second layer of biologically compatible repair fabric to the second surface of the first layer to secure the plurality of preformed tissue grips to the first layer.

2. The method according to claim 1, wherein act (c) includes capturing the grip base between the first layer and the second layer.

3. The method according to claim 1, wherein each preformed tissue grip includes a grip head having a cruciform configuration.

4. The method according to claim 1, wherein the first surface has an outwardly facing curvature and the second surface has an inwardly facing curvature.

5. The method according to claim 1, wherein the plurality of preformed tissue grips are integrated with the second layer.

6. The method according to claim 5, wherein the plurality of preformed tissue grips are molded to the second layer.

7. The method according to claim 1, wherein act (b) includes distributing the plurality of preformed tissue grips in a plurality of distinct grip regions.

8. The method according to claim 7, wherein each grip region includes multiple preformed tissue grips.

9. The method according to claim 7, wherein at least one grip region has a triangular arrangement of preformed tissue grips or a semi-annular arrangement of preformed tissue grips.

10. The method according to claim 7, wherein each grip region has a triangular arrangement of preformed tissue grips.

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